

5

Wind-Induced Vibration of Stay Cables

5.1 Preview

With widespread popularity of cable-stayed bridges around the world, the bridges that have longer spans – over 1000 m – have been constructed by employing increasingly longer stay cables, some over 500 m long. The stay cables are laterally flexible structural members with very low fundamental frequency and very little inherent damping. For this reason, they have been found to be susceptible to various types of vibrations, including wind-induced vibration, rain-wind-induced vibration (RWIV) and deck-induced vibration. These vibrations, as well as other types that have been observed but not effectively categorized, are of significant concern to bridge engineers because they can reach large amplitude and cause unexpected stresses both in the stay cables and in the connections between the cables and other bridge components, potentially leading to strength or fatigue failure of stay cables and other structural members. Recognition of this susceptibility of stay cables leads to the use of various vibration mitigation measures aerodynamically or mechanically. In consideration of the scope of this book, only wind-induced vibration and RWIV of stay cables are discussed in this chapter, whereas vibration mitigation measures for stay cables will be discussed in Chapter 12, together with those for bridge decks and towers.

This chapter will first introduce fundamentals of cable dynamics to lay down a foundation for better understanding of wind-induced vibration and RWIV of stay cables. Wind-induced vibration mainly refers to buffeting by wind turbulence, vortex-induced vibration, galloping of dry inclined cables and wake galloping for groups of cables. A possible mechanism of RWIV will then be discussed and a simple analytical model for RWIV will be presented. The analytical model, together with the fundamentals of cable dynamics, will be applied to predict the RWIV of full-scale stay cables. In order to foresee whether RWIV will occur for a particular cable-stayed bridge at a particular bridge site, a framework for predicting occurrence probability of RWIV will be finally presented.

5.2 Fundamentals of Cable Dynamics

The study of cable dynamics has enjoyed a long and rich history and it is still a popular topic for contemporary research. This section reviews some of basic, but important, concepts in cable dynamics that are relevant to the problem of wind-induced vibration and RWIV of stay cables. It serves as the

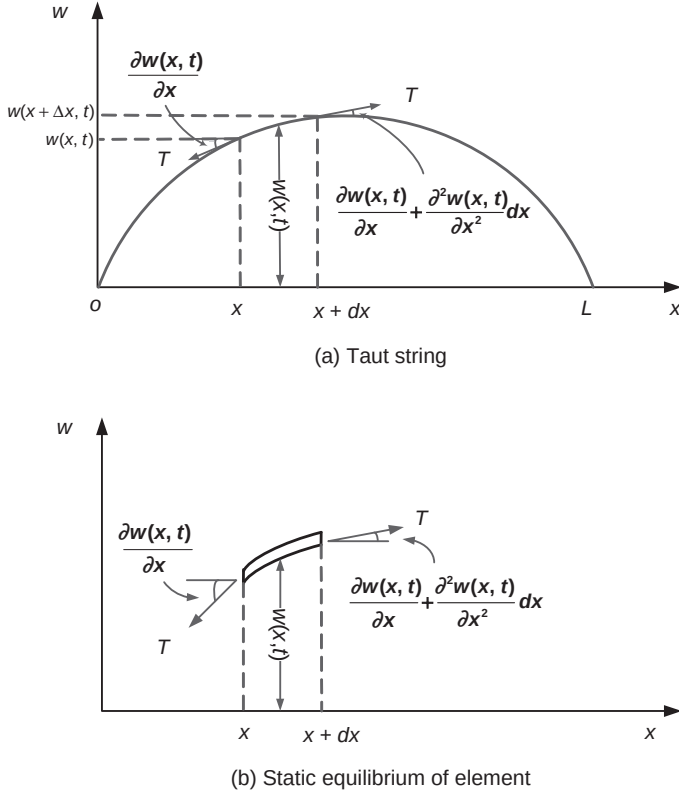


Figure 5.1 A taut string and its element.

background for the discussions in the subsequent sections. The review on this topic is based on those presented in Meirovitch [1] and Zuo [2] for vibration of a taut string, and in Irvine [3] and Zuo [2] for vibration of an inclined cable with sag.

5.2.1 Vibration of a Taut String

Vibration of a taut string is the simplest model for stay cable vibration, which neglects the bending stiffness of the cable and the sag due to self weight. Figure 5.1a shows a taut string, while Figure 5.1b displays an element of the taut string in dynamic equilibrium.

If the displacement w is small, the dynamic equilibrium of the element requires:

$$m \frac{\partial^2 w}{\partial t^2} - T \frac{\partial^2 w}{\partial x^2} + c \frac{\partial w}{\partial t} + F(x, t) = 0 \quad (5.1)$$

with the boundary conditions $w(x, t) = w(L, t) = 0$.

In Equation 5.1: $w(x, t)$ is the in-plane transverse displacement normal to the cable axis at position x ;
 m is the cable mass per unit length;

T is the cable tension force with a constant value;
 c is the internal cable damping coefficient;
 $F(x, t)$ is the wind-induced force on the cable;
 L is the length of the cable.

For an undamped free vibration of the taut string, Equation 5.1 can be reduced to:

$$m \frac{\partial^2 w}{\partial t^2} = T \frac{\partial^2 w}{\partial x^2} \quad (5.2)$$

Assuming that the displacement $w(x, t)$ is in the form of:

$$w(x, t) = W(x)f(t) \quad (5.3)$$

where $f(t)$ is a harmonic function of time with frequency ω , Equation 5.2 becomes:

$$T \frac{\partial^2 W(x)}{\partial x^2} = -\omega^2 m W(x) \quad (5.4)$$

The characteristic equation of Equation 5.4 with the given boundary conditions can be expressed as:

$$\sin\left(\omega L \sqrt{\frac{m}{T}}\right) = 0 \quad (5.5)$$

to which the solution is an infinite number of natural frequencies of the taut string in the form of:

$$\omega_n = \frac{n\pi}{L} \sqrt{\frac{T}{m}}, \quad n = 1, 2, \dots \quad (5.6)$$

with corresponding mode shapes:

$$W_n(x) = A_n \sin \frac{n\pi x}{L}, \quad n = 1, 2, \dots \quad (5.7)$$

where A_n is a constant. The vibration of the taut string can be composed in terms of these natural mode shapes. The undamped free vibration of the taut string can then be completely represented as:

$$w(x, t) = \sum_{i=1}^n A_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{n\pi}{L} \sqrt{\frac{T}{m}} t\right) \quad (5.8)$$

According to Equation 5.8, the different modes of vibration of a taut string is uncoupled. Therefore, the individual modes can be treated as single degree of freedom (SDOF) oscillators.

5.2.2 Vibration of an Inclined Cable with Sag

When the self weight of a stay cable cannot be ignored, the cable develops continuous sag along its chord due to gravity, and thus can no longer be treated as a taut string. The existence of the sag makes the dynamic behavior of the cable in the gravitational plane different from that in the lateral plane and, depending on the magnitude of the sag, this can potentially introduce various kinds of non-linear

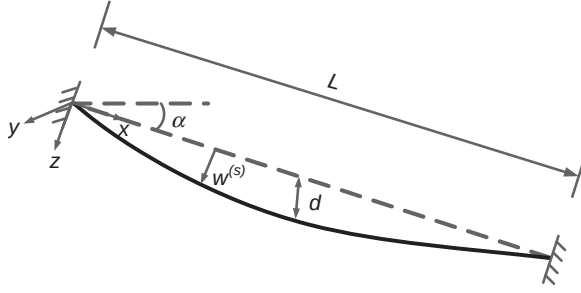


Figure 5.2 Static profile of an inclined, sagged cable.

phenomena [4]. For the stay cables in cable-stayed bridges, the sag is usually very small, with the sag-to-span ratio of the order of 0.001 [4]. As a result, the effect of sag is usually not very significant in stay cable vibrations except in some specific cases, when the vibration is induced by the large amplitude oscillation of the bridge deck [5]. Nevertheless, it is essential to investigate the effects of cable sag, because super long-span cable-stayed bridges with very long stay cables are emerging.

Figure 5.2 depicts the static profile of an inclined, sagged cable in the Cartesian coordinate (x, y, z) . The mass of the cable is uniformly distributed with density m ; the chord length is L ; the sag and chord inclination angle are d and α , respectively. For a metallic cable, the weight to tension ratio and the static strain are relatively small. As a result, the static configuration of the cable can be assumed to be parabolic in the gravitational plane [3].

$$w^s(x) = \frac{1}{2} (mg \cos \alpha L^2 / T) \left[(x/L) - (x/L)^2 \right] \quad (5.9a)$$

$$u^s(x) = v^s(x) = 0 \quad (5.9b)$$

where:

u , v and w are the axial, out-of-plane transverse and in-plane transverse displacements of the cable, respectively;

g is the gravitational constant;

the superscript, s , denotes the static equilibrium state of the cable.

The equation of motion of the cable with sag can be expressed, with respect to its static equilibrium position, as [3]:

$$\frac{\partial}{\partial x} \left[(T + \tau) \frac{\partial u}{\partial x} + \tau \right] + F_x(x, t) = m \frac{\partial^2 u}{\partial t^2} + c_x \frac{\partial u}{\partial t} \quad (5.10a)$$

$$\frac{\partial}{\partial x} \left[(T + \tau) \frac{\partial v}{\partial x} \right] + F_y(x, t) = m \frac{\partial^2 v}{\partial t^2} + c_y \frac{\partial v}{\partial t} \quad (5.10b)$$

$$\frac{\partial}{\partial x} \left[(T + \tau) \frac{\partial w}{\partial x} + \tau \frac{dw^s}{dx} \right] + F_z(x, t) = m \frac{\partial^2 w}{\partial t^2} + c_z \frac{\partial w}{\partial t} \quad (5.10c)$$

where:

τ is the additional dynamic tension due to the cable vibration;

$F_x(x, t)$, $F_y(x, t)$, and $F_z(x, t)$ are the wind-induced forces in the x -, y - and z - directions, respectively;

c_x , c_y , and c_z are the internal cable damping coefficients in the x -, y - and z -directions, respectively.

Assume that the displacement of the cable is small and the stress is still in its elastic range. The dynamic tension can be expressed as:

$$\tau(x, t) = EA\varepsilon(x, t) \quad (5.11)$$

where:

E is the Young's modulus of the cable;

A is the area of the cross-section of the cable;

ε is the dynamic strain, which can be expressed as a function of the displacement as:

$$\varepsilon(x, t) = \frac{\partial u}{\partial x} + \frac{dw^s}{dx} \frac{\partial w}{\partial x} + \frac{1}{2} \left(\frac{\partial u}{\partial x} \right)^2 + \frac{1}{2} \left(\frac{\partial v}{\partial x} \right)^2 + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2 \quad (5.12)$$

If the displacement is small, the second-order terms can be dropped and Equation 5.12 can be reduced to:

$$\varepsilon(x, t) = \frac{\partial u}{\partial x} + \frac{dw^s}{dx} \frac{\partial w}{\partial x} \quad (5.13)$$

For a tightly-stressed cable such as the stay cables in cable-stayed bridges, the axial motion is usually considered unimportant. As a result, Equation 5.10a can be ignored. For the problem of undamped free vibration, the linearized forms of Equations 5.10b and 5.10c reduce to

$$T \frac{\partial^2 v}{\partial x^2} = m \frac{\partial^2 v}{\partial t^2} \quad (5.14a)$$

$$T \frac{\partial^2 w}{\partial x^2} + \tau \frac{d^2 w^s}{dx^2} = m \frac{\partial^2 w}{\partial t^2} \quad (5.14b)$$

Note that Equation 5.14a is essentially the same as Equation 5.2. This suggests that the dynamic behavior of an inclined, sagged cable in the out-of-plane vibration is the same as that of a taut string, which also means that the natural frequencies and associated mode shapes of the out-of-plane vibration are the same as those expressed by Equations 5.6 and 5.7. For the asymmetric modes of the in-plane vibration, the additional dynamic tension is equal to zero and Equation 5.14b can also be reduced to the same as Equation 5.2 or Equation 5.14a. This indicates that the natural frequencies and mode shapes associated with the asymmetric modes of the in-plane vibration are the same as those expressed by Equations 5.6 and 5.7.

In the case of symmetric in-plane modes, additional dynamic tension is not equal to zero, and therefore the natural frequencies and mode shapes of the symmetric modes are different from the corresponding out-of-plane modes. The natural frequencies of the symmetrical in-plane modes can be found by solving the transcendental equation as follows [3]:

$$\tan \frac{\varpi}{2} = \frac{\varpi}{2} - \frac{4}{\lambda^2} \left(\frac{\varpi}{2} \right)^3 \quad (5.15)$$

where the normalized circle frequency ϖ is:

$$\varpi = \omega L \sqrt{\frac{m}{T}} \quad (5.16)$$

and the sag parameters λ^2 is defined as the ratio of the elastic-to-centenary stiffness:

$$\lambda^2 = \left(\frac{mgL}{T} \right)^2 \frac{EA}{T} \quad (5.17)$$

Equation 5.15 is valid for cables of considerable sag-to-span ratio. For cables of small sag, such as the stay cables in cable-stayed bridges, Warnitchai *et al.* [6] suggests that the mode shapes of the symmetric in-plane modes can also be approximately treated as sinusoidal. Therefore, when the sag-to-span ratio and the displacement of the cable both are small, the effects of geometric and material non-linearities can be ignored and, as a result, the dynamic properties of an inclined, sagged cable are very similar to those of a taut string.

Note that the above analysis is based on assumptions that the sag-to-span ratio and the displacement of the cable are both small. Without these two assumptions, geometric and material non-linearities need to be considered, leading to various complicated vibration phenomena, as discussed by many researchers [7–10].

5.3 Wind-Induced Cable Vibrations

There are a number of mechanisms that can potentially lead to wind-induced cable vibrations. This section reviews several wind-induced cable vibrations that require careful consideration by bridge designers. These are: buffeting by wind turbulence; vortex-induced vibration; galloping of dry inclined cable; and wake galloping for groups of cables. The review on this topic follows that presented by FHWA [11].

5.3.1 Buffeting by Wind Turbulence

Flexible structures such as long stay cables undergo substantial motions in strong winds simply because of the random buffeting action of wind turbulence. Very long stay cables will have their lower modes of vibration excited by this effect, but it is not an aeroelastic instability; even very aerodynamically stable structures can be seen to move in strong winds if they are flexible. Buffeting motions are not typically a problem for stay cables in cable-stayed bridges. The buffeting motions increase gradually with wind speed, rather than in the sudden fashion associated with an instability [11].

5.3.2 Vortex-Induced Vibration

Vortex-induced vibration is a classical wind-induced vibration, characterized by limited-amplitude vibrations at relatively low wind speeds. Vortex-induced vibration of a stay cable is caused by the alternate shedding of vortices from the two sides of the cable when the wind is approximately perpendicular to the cable axis. The wind velocity at which the vortex excitation frequency matches the natural frequency of the cable can be found in terms of the Strouhal number:

$$U = \frac{fD}{S} \quad (5.18)$$

where:

U is the mean wind speed;

f is the natural frequency of the cable;

D is the cable diameter;

S is the Strouhal number.

For circular cross-section cables in the Reynolds number range, 10^4 to about 3×10^5 , S is about 0.2.

The amplitude of the cable oscillations is inversely proportional to the Scruton number S_c . It can be seen from $S_c = m\zeta/\rho D^2$ that increasing the mass per unit length m and damping ratio ζ of the cable increases the Scruton number and therefore reduces oscillation amplitudes.

A realistic estimate of inherent cable damping ratios on in-service bridges is in the range from 0.001 to 0.005. For example, a cable consisting of steel strands grouted inside the cable pipe and with a damping ratio of 0.005 has a Scruton number of about 12, and the amplitude of oscillation is only about 0.5% of the cable diameter [11]. During construction and before grouting, the damping ratio of stay cables can be extremely low (e.g. 0.001), and the amplitude could conceivably increase to about 4% of the cable diameter, which is still small [11]. Therefore, vortex shedding from the cables is unlikely to be a major vibration problem for cable-stayed bridges. By adding a small amount of damping, vortex excitation will be suppressed effectively.

5.3.3 Galloping of Dry Inclined Cables

Single cables of circular cross-section do not gallop when they are aligned normal to the wind. However, when the wind velocity has a component that is not normal to the cable axis, an instability with the same characteristics as galloping has been observed; for a single inclined cable, the wind actually acts on an elliptical cross-section of cable [11].

Saito *et al.* [12] conducted a series of wind tunnel experiments on a section of bridge cable mounted on a spring suspension system. Their data suggest an instability criterion given approximately by the following:

$$\left(\frac{U}{fD}\right)_{crit} = 40\sqrt{S_c} \quad (5.19)$$

This formula was for cases where the angle between the cable axis and wind direction was 30° to 60° . The above criterion is a difficult condition to satisfy, particularly for the longer cables of cable-stayed bridges with a typical diameter of 150–200 mm. It was apparent that galloping of dry inclined cables presented the biggest concern and biggest unknown for wind-induced vibration. Further experimental research was necessary to confirm the results of Saito *et al.* [12]. Furthermore, all of their experiments used low levels of damping, so it was important to investigate whether galloping of an inclined cable is possible at damping ratios of 0.005 and higher.

New sets of wind tunnel tests were carried out and the results were reported by FHWA [11]. Testing was performed for various levels of structural damping, cable frequency ratios, surface roughness and at various angles of wind flow. The cable model orientation was changed against the mean wind flow direction for several configurations. Limited-amplitude oscillations were observed under a variety of conditions. The limited-amplitude vibrations occurred within a narrow wind speed range only, which is characteristic of vortex excitation of the high-speed type described by Matsumoto [13].

For the typical cable diameters and wind speeds of concern on cable-stayed bridges, the Reynolds number is in the critical range, i.e. where large changes in the airflow patterns around the cable occur for relatively small changes in the Reynolds number. The excitation mechanism is thus likely to be linked with these changes. The maximum amplitude of the response depended on the orientation angle of the cable.

The results of the new sets of tests obtained by FHWA [11] showed a deviation from the criterion presented by Saito *et al.* [12]. While significant oscillations of the cable occurred (double amplitude up to 1D), it is not conclusive that this was dry inclined cable galloping. In fact, as indicated above, the oscillations had similar characteristics to high speed vortex excitation as discovered by Matsumoto [13]. Divergent oscillations only occurred for one test setup at very low damping, and the vibration had to be suppressed since the setup only allowed for amplitude of 1D. Large vibrations were found only at the lowest damping ratios less than 0.001. Above a damping ratio of 0.003, no significant vibration larger than 10 mm was observed. It can thus be concluded that, if even a low amount of structural damping greater than 0.003 is provided, then vortex shedding and inclined cable galloping vibrations are not significant. This damping corresponds to a Scruton number of approximately 3 [11]. Therefore, dry cable instability should be suppressed by default if enough damping is provided.

5.3.4 Wake Galloping for Groups of Cables

Wake galloping is the elliptical movement caused by variations in drag and acrosswind forces for cables in the wake of other structural components, such as towers or other cables. This occurs at high wind speeds and leads to large amplitude oscillations. These oscillations have been found to cause fatigue of the outer strands of bridge hangers at end clamps on suspension and arch bridges. Similar fatigue problems are a theoretical possibility on cable-stayed bridges but, to date, none have been documented.

The Scruton number is an important parameter with regard to wake galloping effects. An approximate equation for the minimum wind velocity U_{crit} , above which instability can be expected due to wake galloping effects, has been proposed as follows [14,15]:

$$U_{crit} = cfD\sqrt{S_c} \quad (5.20)$$

For circular sections, the constant c has an approximate median value of 40. For cable-stayed bridges, this constant depends on the clear spacing between cables: $c = 25$ for closely spaced cables (2D to 6D spacing); and $c = 80$ for normally spaced cables (generally 10D and higher) [11]. Due to the level of uncertainty associated with practical applications, it is recommended that these values be applied conservatively, exercising engineering judgment.

The critical wind velocity may be low enough to occur commonly during the life of the bridge. Wake galloping, therefore, has the potential to cause serviceability problems. The equation for U_{crit} suggests several possibilities for mitigation. By increasing the Scruton number or natural frequency, the cables will be stable up to a higher wind velocity. However, increasing the frequency is far more effective in raising U_{crit} , due to the square root manifestation of S_c in Equation 5.20. The Scruton number increases with additional damping. The natural frequency may be increased by installing spacers or cross-ties along the cables to shorten the effective length of cable for the vibration mode of concern.

It should be noted that wake galloping is not a major design concern for normal, well-separated cable arrangements. For unusual cases, however, it is recommended that some attention be paid to the possibility of wake galloping.

5.4 Mechanism of Rain-Wind-Induced Cable Vibration

5.4.1 Background

The combination of rain and moderate wind speeds can cause high-amplitude cable vibrations at low frequencies. This phenomenon has been observed on many cable-stayed bridges and has been studied in detail [11].

Rain-wind-induced vibrations (RWIV) were first identified by Hikami and Shiraishi on the Meiko-Nishi cable-stayed bridge [16]. Since then, these vibrations have been observed on other cable-stayed bridges around the world [17,18]. Large amplitude cable vibrations under simultaneous occurrence of wind and rain may induce undue stresses in the cable in the vicinity of the anchorage, potentially leading to fatigue fractures of internal wire strands [19]. Sometimes they may also damage the steel tubes installed on the anchorage for protecting the cable [20].

Many studies have thus been conducted to reveal the mechanism and conditions of RWIV of stay cables. These studies include field measurements of RWIV of stay cables [16,21], wind-rain tunnel tests of sectional cable models [22–24] and analytical studies for exploring the mechanism of wind-rain-induced cable vibration [23,25–27]. It was found that these vibrations occurred typically when there was rain and moderate wind speeds (8–15 m/s) in the direction angled 20° to 60° to the cable plane, with the cable declined in the direction of the wind. The frequencies were low, typically less than 3 Hz. The peak amplitudes were very high, in the range of 0.25 m to 1.0 m – violent movements, resulting in the clashing of adjacent cables observed in several cases.

Wind-rain tunnel tests have shown that rivulets of water running down the upper and lower surfaces of the cable in rainy weather were the essential component of this aeroelastic instability [16]. The water rivulets changed the effective shape of the cable and moved as the cable oscillated, causing cyclical changes in the aerodynamic forces that led to the wind feeding energy into oscillations. The wind direction causing the excitation was approximately 45° to the cable plane. The particular range of wind velocities that caused the oscillations appears to be that which maintained the upper rivulet within a critical zone on the upper surface of the cable.

Since some bridges have been built without experiencing problems from RWIV of stay cables, it appears probable that, in some cases, the level of damping naturally present is sufficient to avoid the problem. The rig test data of Saito *et al.* [12], obtained using realistic cable mass and damping values, are useful in helping to define the boundary of instability for rain and wind oscillations. Based on their results, it appears that rain-wind oscillations can be reduced to a harmless level if the Scruton number is greater than 10 [15].

RWIV are highly problematic vibrations, with their large amplitudes and relatively frequent occurrence, and this is among the most significant considerations in the design of mitigation measures for stay cables. One of the primary research components is to acquire a more in-depth understanding of the underlying mechanics of rain-wind-induced stay cable vibration and to develop an analytical model with the capability to predict major characteristics of RWIV for an arbitrary stay cable. Such a model will enable a much more comprehensive treatment of the problem, both in terms of when such vibrations are likely to occur and what will be the effect of various mitigation approaches. An initial attempt at solving such a problem is presented in the following sections for reference [27–29].

5.4.2 Analytical Model of SDOF

Let us use a rigid and uniform inclined cylinder to represent a stay cable segment (see Figure 5.3). The inclination of the cylinder is denoted by angle α , and the yaw angle of the incident wind is designated by angle β . The cylinder is supposed to be supported by springs at its ends in the plane 1-5-7. The consideration of such a cylinder, rather than a real cable, is because many researchers used it in their rain-wind simulation tests [16,30], and some of the test results have to be used in this section for exploration of mechanism of RWIV.

As a preliminary analytical study, the upper rivulet is assumed to distribute uniformly along the longitudinal axis of the cylinder and vibrate circumferentially over the surface of the cylinder. Turbulent effect and axial flow effect are not considered. The static position of the upper rivulet due to the mean wind when the cylinder is stationary is defined by angle θ_0 , and the dynamic angular displacement of the rivulet as the cylinder vibrates is designated by θ with reference to θ_0 (see Figure 5.4).

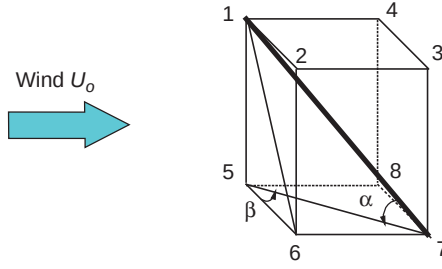


Figure 5.3 Orientation of inclined cylinder (from [27]) (Reproduced with permission from Elsevier).

Since the cylinder is not perpendicular to the direction of the mean wind speed U_0 , one needs to find the component of mean wind speed perpendicular to the cylinder, U , using the following equation.

$$U = U_0 \sqrt{\cos^2 \beta + \sin^2 \alpha \sin^2 \beta} = U_0 \sqrt{\sin^2 \alpha + \cos^2 \alpha \cos^2 \beta} \quad (5.21)$$

The angle of attack of the mean wind speed component U is defined as γ (see Figure 5.4), which indicates the stagnation point of incident wind on the surface of cylinder. The position of the stagnation point depends on many factors, such as the cross-section of cylinder with rivulet, roughness of cylinder surface, wind turbulence and cable motion. No experimental results are available to the author at this stage with respect to the position of stagnation point on a stay cable with rivulet. Thus, the angle of attack in this study is selected as the ideal angle of attack multiplied by an influence factor ε .

$$\gamma = \varepsilon \sin^{-1} \left(\frac{\sin \alpha \sin \beta}{\sqrt{\cos^2 \beta + \sin^2 \alpha \sin^2 \beta}} \right) \quad (5.22)$$

When ε is set as 1, γ represents the ideal angle of attack for the cylinder without rivulet [31]. When ε is selected as zero, it indicates that the position of the stagnation point is the same as that on the

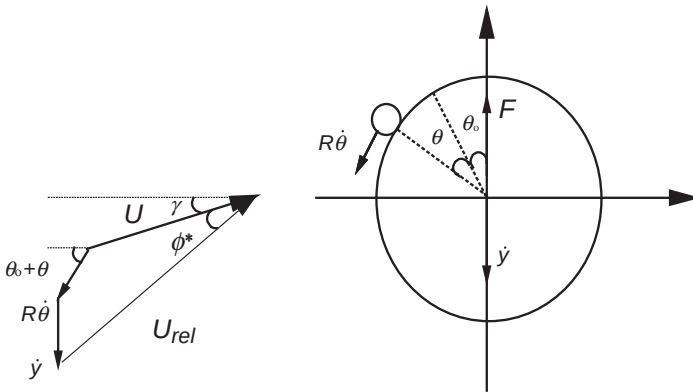


Figure 5.4 Relative velocity to cylinder and moving rivulet (from [27]) (Reproduced with permission from Elsevier).

cylinder without rivulet and yaw angle. The effects of the mean wind speed component along the cylinder axis and wind turbulence are not considered in this study.

In consideration of the transverse vibration of the cable of velocity $\dot{y}(t)$ and the angular vibration of the upper rivulet of velocity $\dot{\theta}(t)$, the relative velocity of mean wind to the cylinder with moving rivulet is, therefore:

$$U_{rel} = \sqrt{(U \cos \gamma + R\dot{\theta} \cos(\theta + \theta_0))^2 + (U \sin \gamma + \dot{y} + R\dot{\theta} \sin(\theta + \theta_0))^2} \quad (5.23)$$

The angle between the relative velocity U_{rel} and the horizontal axis in Figure 5.4 is defined as $\varphi^*(t)$, which can be expressed as:

$$\varphi^* = \tan^{-1} \frac{U \sin \gamma + \dot{y} + R\dot{\theta} \sin(\theta + \theta_0)}{U \cos \gamma + R\dot{\theta} \cos(\theta + \theta_0)} \quad (5.24)$$

where R is the radius of the cylinder and the size of the rivulet is considered small compared with the diameter of the cylinder. The field and laboratory observations indicate that $R\dot{\theta}$ is very small compared with the concerned mean wind speed U . Thus, if the angles γ and $\varphi^*(t)$ are limited a certain range, Equation 5.24 can be reduced as

$$\begin{aligned} \varphi^* &\approx \frac{U \sin \gamma + \dot{y} + R\dot{\theta} \sin(\theta + \theta_0)}{U \cos \gamma} \\ &\approx \frac{U \sin \gamma + \dot{y} + R\dot{\theta} \left[(\theta + \theta_0) - \frac{1}{6}(\theta + \theta_0)^3 \right]}{U \cos \gamma} \end{aligned} \quad (5.25)$$

The net vertical force on the cylinder per unit length in the y direction is then:

$$F = \frac{\rho D U_{rel}^2}{2} [C_L(\varphi) \cos \varphi^* + C_d(\varphi) \sin \varphi^*] \quad (5.26)$$

where:

ρ is the density of the air;

D is the diameter of the cross-section of the cylinder;

C_d is the drag coefficient;

C_L is the lift coefficient.

The drag and lift coefficients of the cylinder with rigid rivulet measured from the wind tunnel tests are often expressed as the function of the angle φ defined in Figure 5.5 [16,32]. The relationship between the angles φ^* and φ is given by:

$$\varphi = \varphi^* - \theta - \theta_0 = \frac{U \sin \gamma + \dot{y} + R\dot{\theta} \left[(\theta + \theta_0) - \frac{1}{6}(\theta + \theta_0)^3 \right]}{U \cos \gamma} - \theta - \theta_0 \quad (5.27)$$

In consideration that the cylinder structural damping is viscous and the cylinder mass is uniformly distributed, the equation of vertical motion of the cylinder can be written as:

$$\ddot{y} + \omega_c^2 y + 2\xi_c \omega_c \dot{y} + \frac{F}{m} = 0 \quad (5.28)$$

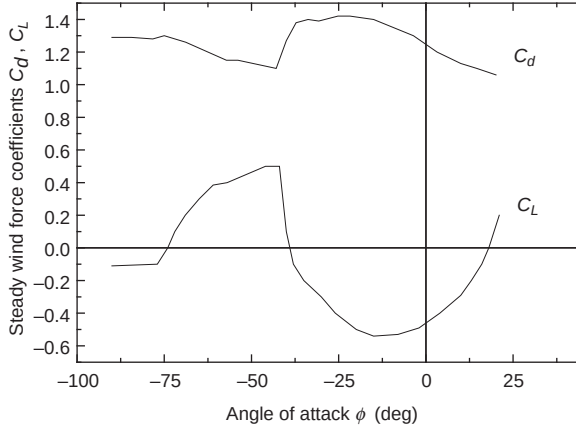


Figure 5.5 Aerodynamic coefficients vs. wind angle of attack (from [27]) (Reproduced with permission from Elsevier).

where:

ω_c is the circular natural frequency of the cylinder;

ξ_c is the structural damping ratio of the cylinder;

m is the mass of the cylinder per unit length.

Now fit the drag coefficient and lift coefficient in Equation 5.26 with respect to φ using the first three terms of Taylor's series:

$$C_L(\varphi) = A_0 + A_1\varphi + \frac{A_2}{2}\varphi^2 + \frac{A_3}{6}\varphi^3 \quad (5.29a)$$

$$C_d(\varphi) = B_0 + B_1\varphi + \frac{B_2}{2}\varphi^2 + \frac{B_3}{6}\varphi^3 \quad (5.29b)$$

The substitution of Equations 5.29a and 5.29b into Equation 5.26 and then the expansion of the sine and cosine functions in Equation 5.26 with respect to φ^* and the use of the first two terms of the power series yield the vertical force acting on the cylinder per unit length of the form:

$$F = \frac{\rho D U_{rel}^2}{2} \left[\left(A_0 + A_1\varphi + \frac{1}{2}A_2\varphi^2 + \frac{1}{6}A_3\varphi^3 \right) \left(1 - \frac{\varphi^{*2}}{2} \right) + \left(B_0 + B_1\varphi + \frac{1}{2}B_2\varphi^2 + \frac{1}{6}B_3\varphi^3 \right) \left(\varphi^* - \frac{\varphi^{*3}}{6} \right) \right] \quad (5.30)$$

The substitution of Equations 5.21, 5.23, 5.25 and 5.27 into Equation 5.30 and then the reservation of the linear terms of $\dot{y}$, θ and $\dot{\theta}$ only lead to

$$F = \frac{\rho D}{2} (\Gamma_1 R U_0 \dot{\theta} + \Gamma_2 U_0 \dot{y} + \Gamma_3 U_0^2 \theta) \quad (5.31)$$

where Γ_1 , Γ_2 , Γ_3 are the rain-wind aerodynamic functions. They are not only the function of the cable inclination, the wind yaw angle, the wind stagnation point and the mean wind speed via the static position of rivulet, but also the function of cable motion and rivulet motion, which may change the values of drag coefficient and lift coefficient.

$$\begin{aligned}
\Gamma_1 = & \left[c_0 c_4 \left(1 - \frac{c_5^2}{2} \right) + c_1 c_5 \left(1 - \frac{c_5^2}{6} \right) \right] B_0 + \left[c_0 c_4 c_5 \left(1 - \frac{c_5^2}{6} \right) + c_0 c_4 c_6 \left(1 - \frac{c_5^2}{2} \right) + c_1 c_5 c_6 \left(1 - \frac{c_5^2}{6} \right) \right] B_1 \\
& + \left[c_0 c_4 c_5 c_6 \left(1 - \frac{c_5^2}{6} \right) + \frac{c_0 c_4 c_6^2}{2} \left(1 - \frac{c_5^2}{2} \right) + \frac{c_1 c_5 c_6^2}{2} \left(1 - \frac{c_5^2}{6} \right) \right] B_2 \\
& + \left[\frac{c_0 c_4 c_5 c_6^2}{2} \left(1 - \frac{c_5^2}{6} \right) + \frac{c_0 c_4 c_6^3}{6} \left(1 - \frac{c_5^2}{2} \right) + \frac{c_1 c_5 c_6^3}{6} \left(1 - \frac{c_5^2}{6} \right) \right] B_3 \\
& + \left[c_1 \left(1 - \frac{c_5^2}{2} \right) - c_0 c_4 c_5 \right] A_0 + \left[c_0 c_4 \left(1 - \frac{c_5^2}{2} \right) + c_1 c_6 \left(1 - \frac{c_5^2}{2} \right) - c_0 c_4 c_5 c_6 \right] A_1 \\
& + \left[c_0 c_4 c_6 \left(1 - \frac{c_5^2}{2} \right) - \frac{c_0 c_4 c_5 c_6^2}{2} + \frac{c_1 c_6^2}{2} \left(1 - \frac{c_5^2}{2} \right) \right] A_2 \\
& + \left[\frac{c_0 c_4 c_6^2}{2} \left(1 - \frac{c_5^2}{2} \right) - \frac{c_0 c_4 c_5 c_6^3}{6} + \frac{c_1 c_6^3}{6} \left(1 - \frac{c_5^2}{2} \right) \right] A_3
\end{aligned} \tag{5.32a}$$

$$\begin{aligned}
\Gamma_2 = & \left[c_0 c_3 \left(1 - \frac{c_5^2}{2} \right) + c_2 c_5 \left(1 - \frac{c_5^2}{6} \right) \right] B_0 + \left[c_0 c_3 c_5 \left(1 - \frac{c_5^2}{6} \right) + c_0 c_3 c_6 \left(1 - \frac{c_5^2}{2} \right) + c_2 c_5 c_6 \left(1 - \frac{c_5^2}{6} \right) \right] B_1 \\
& + \left[c_0 c_3 c_5 c_6 \left(1 - \frac{c_5^2}{6} \right) + \frac{c_0 c_3 c_6^2}{2} \left(1 - \frac{c_5^2}{2} \right) + \frac{c_2 c_5 c_6^2}{2} \left(1 - \frac{c_5^2}{6} \right) \right] B_2 \\
& + \left[\frac{c_0 c_3 c_5 c_6^2}{2} \left(1 - \frac{c_5^2}{6} \right) + \frac{c_0 c_3 c_6^3}{6} \left(1 - \frac{c_5^2}{2} \right) + \frac{c_2 c_5 c_6^3}{6} \left(1 - \frac{c_5^2}{6} \right) \right] B_3 \\
& + \left[c_2 \left(1 - \frac{c_5^2}{2} \right) - c_0 c_3 c_5 \right] A_0 + \left[c_0 c_3 \left(1 - \frac{c_5^2}{2} \right) + c_2 c_6 \left(1 - \frac{c_5^2}{2} \right) - c_0 c_3 c_5 c_6 \right] A_1 \\
& + \left[c_0 c_3 c_6 \left(1 - \frac{c_5^2}{2} \right) - \frac{c_0 c_3 c_5 c_6^2}{2} + \frac{c_2 c_6^2}{2} \left(1 - \frac{c_5^2}{2} \right) \right] A_2 \\
& + \left[\frac{c_0 c_3 c_6^2}{2} \left(1 - \frac{c_5^2}{2} \right) - \frac{c_0 c_3 c_5 c_6^3}{6} + \frac{c_2 c_6^3}{6} \left(1 - \frac{c_5^2}{2} \right) \right] A_3
\end{aligned} \tag{5.32b}$$

$$\Gamma_3 = -c_0 c_5 \left(1 - \frac{c_5^2}{6} \right) \left(B_1 + B_2 c_6 + \frac{1}{2} B_3 c_6^2 \right) - c_0 \left(1 - \frac{c_5^2}{2} \right) \left(A_1 + A_2 c_6 + \frac{1}{2} A_3 c_6^2 \right) \tag{5.32c}$$

in which:

$$c_0 = \cos^2 \beta + \sin^2 \alpha \sin^2 \beta \tag{5.33a}$$

$$c_1 = \left[2 - (\gamma - \theta_0)^2 \right] \sqrt{c_0} \tag{5.33b}$$

$$c_2 = 2 \sin \gamma \sqrt{c_0} \tag{5.33c}$$

$$c_3 = \frac{1}{\sqrt{c_0} \cos \gamma} \tag{5.33d}$$

$$c_4 = \frac{\theta_0 - \frac{\theta_0^3}{6}}{\sqrt{c_0} \cos \gamma} \quad (5.33e)$$

$$c_5 = \tan \gamma \quad (5.33f)$$

$$c_6 = c_5 - \theta_0 \quad (5.33g)$$

With the information on the orientation of a stay cable (α , β), the stagnation point influence factor (ε), the static position of upper rivulet related to the mean wind speed (θ_0) and the drag and lift coefficients of a cable segment with artificial rivulet measured from wind tunnel tests (A_i , B_i , $i = 0, 1, 2, 3$), one can readily identify the aerodynamic functions Γ_1 , Γ_2 , Γ_3 .

The combination of Equation 5.28 with Equation 5.31 results in:

$$\ddot{y} + \omega_c^2 y + \left(2\xi_c \omega_c + \frac{\rho D \Gamma_2 U_0}{2m} \right) \dot{y} = -\frac{\rho D}{2m} (\Gamma_1 R U_0 \dot{\theta} + \Gamma_3 U_0^2 \theta) \quad (5.34)$$

Based on the observations from either field measurements or simulated rain-wind tunnel tests [16], the motion of upper rivulet, θ , can be assumed to be harmonic as long as the steady-state vibration is concerned:

$$\theta = a \sin \bar{\omega} t \quad (5.35)$$

The frequency of rivulet motion is almost the same as that of cable motion, as observed from rain-wind tunnel tests when the vibration is in steady state [16]. The amplitude of rivulet motion, a , is also taken from rain-wind tunnel tests. Thus, the motion of the rivulet is assumed as a known function of time in this section. Substituting Equation 5.35 into Equation 5.34 produces:

$$\ddot{y} + \omega_c^2 y + 2\tilde{\xi} \omega_c \dot{y} = -\frac{\rho D a}{2m} (\Gamma_1 R U_0 \bar{\omega} \cos \bar{\omega} t + \Gamma_3 U_0^2 \sin \bar{\omega} t) = \tilde{F}(t) \quad (5.36)$$

in which:

$$\tilde{\xi} = \xi_c + \frac{\rho D \Gamma_2 U_0}{4m \omega} = \xi_c + \xi_a \quad (5.37)$$

where:

$\tilde{\xi}$ is the total damping ratio of the cylinder;

ξ_a is the rain-wind aerodynamic damping ratio;

$\tilde{F}(t)$ is the normalized force due to the motion of rivulet.

Since Γ_2 changes with the mean wind speed U_0 through the static position of rivulet, and depends on the motions of cable and rivulet, the aerodynamic damping ratio and the total damping ratio are a function of time.

5.4.3 Horizontal Cylinder with Fixed Rivulet

There are two approaches currently used to simulate rivulet on a cable section model in wind tunnel simulation tests: one is to spray water appropriately onto the surface of the cable model to form moving rivulet [33]; and the other is to stick artificial rivulet on the cable surface [32]. To start with the simplest case, this section investigates the dynamic behavior of horizontal cylinder with fixed rivulet using the

developed model, then compares the results with the wind tunnel test results obtained by Gu *et al.* [32]. Note that this simplest case does not represent the RWIV of stay cables and it is only for exploring the mechanism of RWIV using the proposed analytical model.

When the rivulet is fixed, the dynamic motion $\theta(t)$ of the rivulet relative to the cylinder is equal to zero and the static position of the rivulet θ_0 is no longer the function of the mean wind speed. The equation of vertical motion of the horizontal cylinder becomes:

$$\ddot{y} + \omega_c^2 y + 2\tilde{\xi}\omega_c \dot{y} = 0 \quad (5.38)$$

Γ_2 in Equation 5.37 is also changed. For instance, Γ_2 can be expressed by the following equation for the horizontal cylinder with zero wind yaw angle β :

$$\Gamma_2 = B_0 - B_1\theta_0 + \frac{B_2}{2}\theta_0^2 - \frac{B_3}{6}\theta_0^3 + A_1 - A_2\theta_0 + \frac{A_3}{2}\theta_0^2 \quad (5.39)$$

For the cylinder with fixed rivulet, the coefficients A_i and B_i ($i = 0, 1, 2, 3$) may change with the angle φ , which, in turn depends on the cylinder velocity $\dot{y}$ and the angle θ_0 . Therefore, Equation 5.38 is a non-linear differential equation. The Runge-Kutta method, using the computer package MATLAB as a platform, is employed to find the solution of the equation of motion of the cylinder with rivulet.

Let us consider a horizontal cylinder of 120 mm diameter with a fixed upper rivulet [32]. The upper rivulet is modeled as a half ellipse, with a long axis of 14.5 mm and a short axis of 10 mm. The measured drag and lift coefficients are plotted in Figure 5.5. It can be seen that when φ is equal to -43° , the derivative of lift coefficient with angle of attack has a sudden change from a positive value to a negative value, whereas the derivative of the drag coefficient changes from a negative value to a positive value. To have the best fit of the measured aerodynamic coefficients using the first three terms of Taylor's series, the measured aerodynamic coefficients are divided into the two ranges, distinguished by the critical angle φ of -43° .

Figure 5.6a shows the time history of displacement response of the horizontal cylinder with fixed rivulet, obtained by using Equation 5.27 and Equations 5.37 to 5.39 with a small initial displacement. The cylinder has the following parameters: the mass per unit length of the cylinder m is 3.48 kg/m; the natural frequency f_n is 1.28 Hz; and the structural damping ratio ξ_s is 0.2%. The mean wind speed U_0 and yaw angle β are 10 m/s and zero, respectively. The air density is 1.225 kg/m³ and the position of the fixed rivulet θ_0 is 32° .

The computed result shows that, after giving a small initial displacement, the vibration amplitude of the cylinder increases with time within the first 320 seconds, because the total damping ratio $\tilde{\xi}$ is negative and constant during this period (see Figure 5.6b). After the amplitude of the cylinder is increased to a certain level, the angle φ reaches the value of -43° . The aerodynamic damping ratio and the total damping ratio then change alternately between a negative value and a positive value, rather than remaining constant (Figure 5.6b). As a result of the alteration of aerodynamic damping, the motion of the cylinder becomes almost periodic, with a nearly constant peak-to-peak-amplitude after 320 seconds.

To know the effect of position of the fixed rivulet θ_0 on cylinder vibration, the motion of the horizontal cylinder with a wind yaw angle of 30° is computed against a series of angles θ_0 . The computed results are plotted in Figure 5.7a, together with the test results [32], in which A_{max} means the maximum peak to peak amplitude. A_{allow} means the allowable peak-to-peak amplitude, which was set as 530 mm in the simulated wind tunnel tests. The mean wind speed U_0 used in the computation is 18 m/s. $A_{max}/A_{allow} = 1$ indicates that the vibration amplitude is larger than the allowable amplitude.

It can be seen that the computed and measured results both demonstrate that, within a certain range of U_0 , the cylinder has large amplitude vibration. However, out of this range, the cylinder has very small vibration, or it stops vibrating very quickly after a small initial displacement or disturbance. The

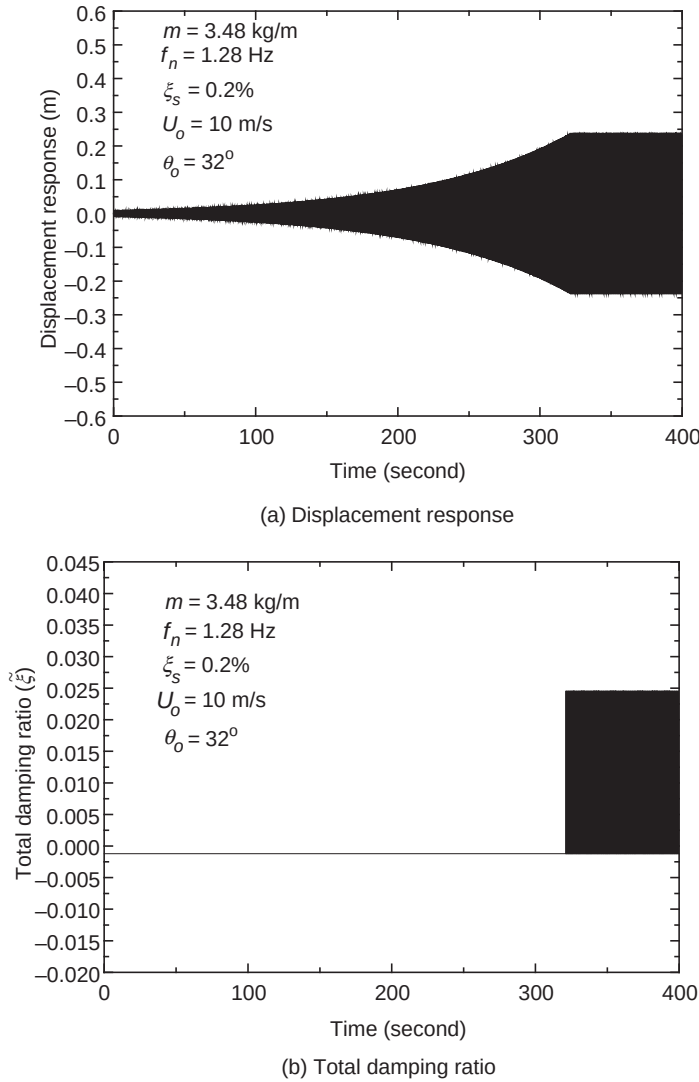


Figure 5.6 Time histories of horizontal cylinder vibration with fixed rivulet (from [27]) (Reproduced with permission from Elsevier).

cylinder starts its large amplitude vibration at 30° , measured by the wind tunnel test and also predicted by the analytical model. There is some difference in the upper bounds of the range between the computed and measured results.

Figure 5.7b shows the variation of the normalized vibration amplitude of the cylinder with mean wind speed for the case of 30° wind yaw angle and 35° static rivulet position. Again, the computed results are in good agreement with the test results. Both the wind tunnel test results and the computed results show a kind of amplitude-restricted vibration. The reason why the vibration amplitude is limited is that when the vibration amplitude of the cylinder reaches a certain level, the aerodynamic damping ratio and the total damping ratio change periodically between a positive peak value and a negative peak value.

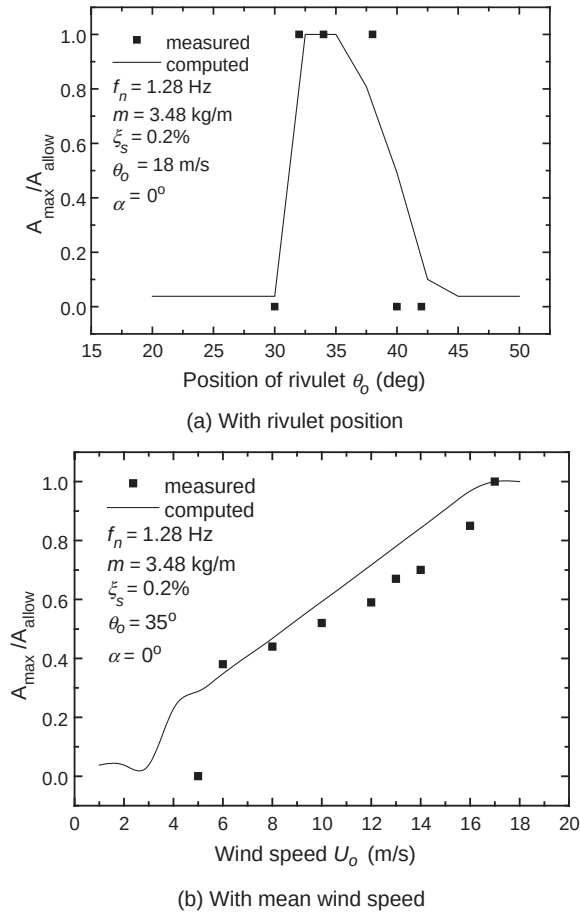


Figure 5.7 Variations of cylinder vibration amplitude (from [27]) (Reproduced with permission from Elsevier).

5.4.4 Inclined Cylinder with Moving Rivulet

To investigate the capability of the analytical model for predicting the motion of inclined cable with moving rivulet, the inclined cylinder tested in a simulated rain-wind tunnel by Hikami and Shiraishi [16] is selected in this study. The diameter of the cylinder is 140 mm; the mass per unit length is 10.2 kg/m; both the inclination and yaw angles are 45° ; and the natural frequency is 1 Hz. The structural damping ratio and the stagnation influence factors are assumed to be 0.7% and 0.4, respectively. The frequency of steady-state upper rivulet motion is the same as the natural frequency of the cylinder [16]. The drag and lift coefficient curves of the cylinder with upper rivulet ($d/D = 0.1$) were reported by Yamaguchi [25] and are reproduced in Figure 5.8a.

It can be seen that there is a sudden change in the gradient of the curves at an angle of -55° . The curves are subsequently fitted to Taylor's series of the first three terms distinguished by the critical angle of -55° . For the moving rivulet, its static position θ_0 is the function of mean wind speed. The wind tunnel test results related to this position [16] are plotted in Figure 5.8b and fitted by a quadratic function.

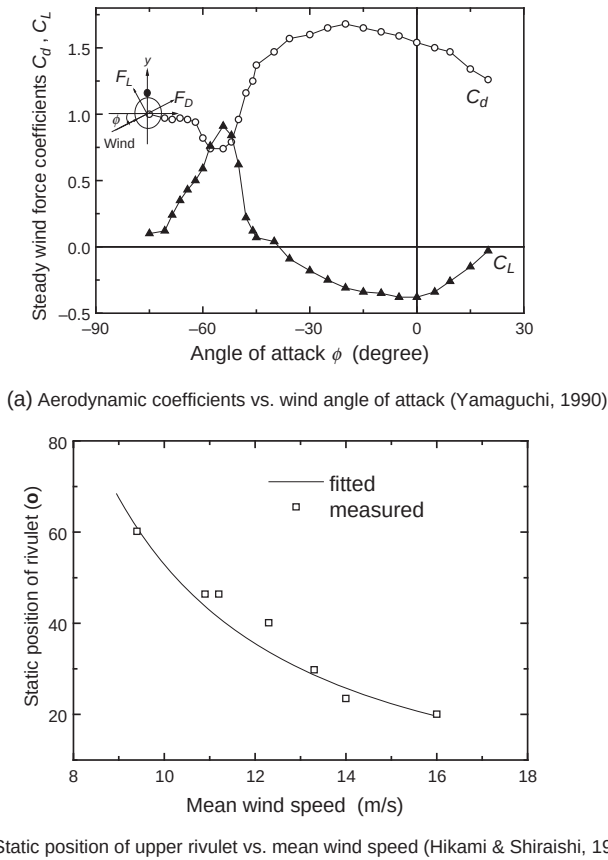


Figure 5.8 Aerodynamic properties of rain-wind-induced cylinder vibration (from [27]) (Reproduced with permission from Elsevier).

The results obtained from the analytical model are shown in Figure 5.9, together with the rain-wind tunnel test results. In this figure, the x -coordinate is the mean wind speed U_0 and the y -coordinate is the maximum displacement response amplitude of the cylinder. Both analytical and measured results show that the rain-wind-induced cylinder vibration only occurs within a certain range of mean wind speed. It is also a vibration of restricted amplitude rather than galloping. The predicted maximum vibration amplitude is moderately larger than that from the rain-wind tunnel tests but the onset mean wind speed at which the large cylinder vibration starts is slightly lower in the computation than in the rain-wind tunnel test. Such differences may be due to uncertainties in the selection of stagnation influence factor and structural damping ratio.

One particular mean wind speed corresponding to the global maximum displacement response, i.e. 9.8 m/s, is selected to examine cable vibration features. It is found that at this mean wind speed, the angle $\phi(t)$ changes with time, and its negative amplitude is less than -55° . Thus, the aerodynamic damping ratio and the total damping ratio alternate with time. Because the structural damping ratio is relatively large, the total damping ratio is kept within the positive value range. However, the upper and lower bounds of the total damping ratio are quite small, but the peak-to-peak amplitude of excitation force due to the motion of rivulet is relatively larger. As a result, the cylinder exhibits a large amplitude vibration. One may thus conclude that the restricted large amplitude vibration of the cylinder at 9.8 m/s

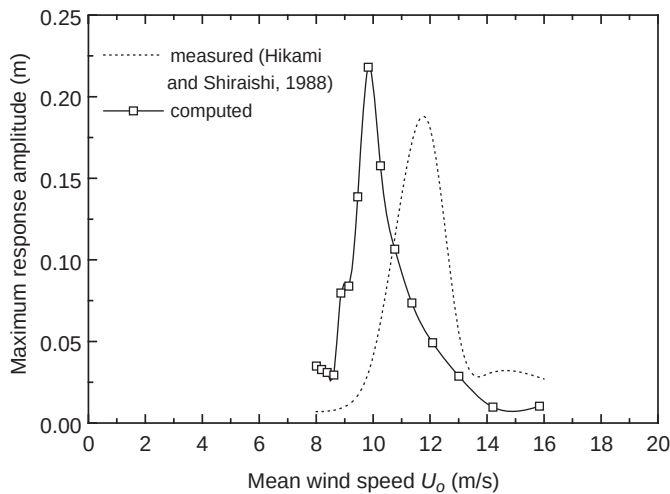


Figure 5.9 Maximum cylinder vibration amplitude vs. mean wind speed (from [27]) (Reproduced with permission from Elsevier).

wind speed is because of a very small total damping ratio and relatively large amplitude force due to the motion of rivulet.

The time-history of the displacement response of the cylinder is shown in Figure 5.10. During the first 50 seconds, the cylinder vibration is in a transient vibration period and the smallest amplitude of the angle $\varphi(t)$ is above -55° . This transient vibration period does not reflect the real situation because of the steady-state rivulet motion assumption used in this study. However, when the vibration amplitude of cylinder increases to a certain level, the smallest amplitude of the angle $\varphi(t)$ becomes less than -55° and the aerodynamic damping ratio and the total damping ratio no longer remain constant. The

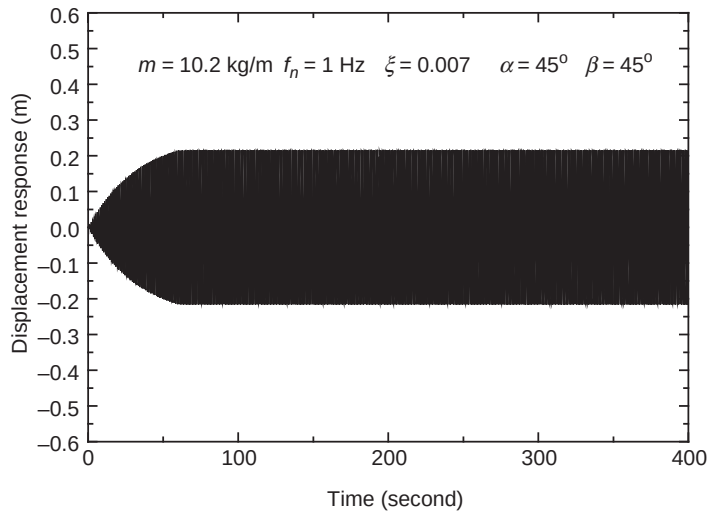


Figure 5.10 Time history of cylinder vibration with moving rivulet (from [27]) (Reproduced with permission from Elsevier).

normalized force also changes with time because of the interaction between incident wind, moving rivulet and cylinder motion. All of these features of rain-wind-induced cable vibration cause the cylinder to vibrate with almost constant amplitude.

5.4.5 Analytical Model of 2DOF

The discussions above show that the SDOF model could capture main features of rain-wind-induced cable vibration, such as velocity-restricted vibration and amplitude-restricted vibration. The occurrence of RWIV of the cylinder was mainly because of alternating aerodynamic damping and/or aerodynamic force due to the interaction between rivulet, cable and wind. However, the motion of rivulet was assumed to be a known harmonic motion in the SDOF model and, accordingly, only the steady-state RWIV could be predicted. In this section, the SDOF model is extended to a 2DOF model by including the equation of motion of the upper rivulet. In the 2DOF model, the rotating motion of the rivulet around the central axis of the cylinder is considered together with the transverse motion of the inclined cylinder. The interaction between the upper rivulet and the cylinder is described in terms of non-linear damping force, linear restoring force and inertia force [29].

For the upper rivulet, it can be assumed that the mean wind force, the supporting force from the cylinder and the gravity force on the rivulet all keep the rivulet in the static position θ_0 . The dynamic equilibrium of the rivulet in the circumferential direction of the cylinder depends on the rivulet inertia force and the interacting force from the cylinder at the contacting surface with the rivulet. As the absolute acceleration of the rivulet is the superimposition of the cylinder vertical acceleration at the contacting point and the acceleration of the rivulet relative to the cylinder, the acceleration of the rivulet in the tangential direction at the contacting point with the cylinder is given by $\ddot{y} \sin(\theta + \theta_0) + R\ddot{\theta}$.

As well as the inertial force, the rivulet is subjected to turbulent wind force, aerodynamic damping force, restoring force due to water surface tension and friction force between water and cylinder surface. The restoring force and the friction force depend on many factors, such as the contacting surface roughness and the size and density of the rivulet. The turbulent wind force and the aerodynamic damping force are also difficult to be quantified. For the sake of simplification, the combined effect of all damping forces on the rivulet is approximately represented by $-c_r |\dot{x}|^{\alpha_r} \dot{x}$. Here, $\dot{x}$ is the relative velocity of the rivulet to the cylinder, which is equal to $R\dot{\theta}$; c_r is the damping coefficient of the rivulet; α_r is a predetermined exponent; $\alpha_r = 0$ represents a viscous damping force; $\alpha_r = 1$ indicates the damping force from an orifice of square law.

The restoring force due to water surface tension is assumed to be a linear function of the relative movement of the rivulet to the cylinder at this stage. The turbulence wind force is ignored, since the size of the rivulet is small and the shape of the rivulet is naturally toward the streamline. Within all the above considerations, the dynamic equation of motion of the rivulet is expressed as:

$$m_r \left(\ddot{\theta} + \frac{\ddot{y} \sin(\theta_0 + \theta)}{R} \right) + c_r |\dot{\theta}|^{\alpha_r} \dot{\theta} + k_r \theta = 0 \quad (5.40)$$

where m_r , c_r , and k_r are the mass, damping coefficient, and stiffness coefficient per unit length of the rivulet. Equation 5.36 and Equation 5.40 can be combined and simplified as follows:

$$\begin{cases} \ddot{y} + 2\tilde{\xi}\omega_c\dot{y} + \omega_c^2 y = \tilde{F} \\ \ddot{\theta} + \frac{\ddot{y} \sin(\theta_0 + \theta)}{R} + 2\tilde{\xi}_r\omega_r |\dot{\theta}|^{\alpha_r} \dot{\theta} + \omega_r^2 \theta = 0 \end{cases} \quad (5.41)$$

Equation 5.41 is a strong non-linear equation. The 4th order Runge-Kutta method can be applied to find the solution of motion for both the cylinder and the rivulet. Figure 5.11 displays the maximum

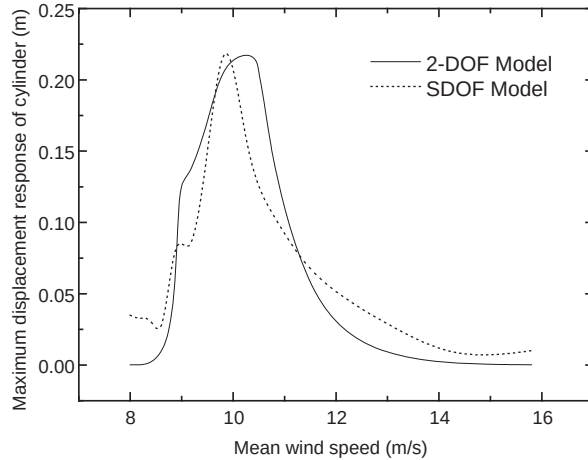


Figure 5.11 Comparison between 2DOF model and SDOF model (from [29]) (Reproduced with permission from Techno.Press).

displacement response of the cylinder against mean wind speed obtained by both the SDOF model and the 2DOF model. The parameters of the cylinder are the same for both the computed cases.

It can be seen that, at very low and high mean wind speeds, the vibration amplitude of the cylinder predicted by the SDOF model is slightly larger than that predicted by the 2DOF model. Within the range around the critical mean wind speed, the vibration amplitude of the cylinder predicted by the 2DOF model is slightly larger than that computed by the SDOF model. In comparison with the measured results shown in Figure 5.9, the computed curve of vibration amplitude of the cylinder by the 2DOF model is slightly closer to the measured results than that computed from the SDOF model. It seems that the two degrees of freedom model is a better model to predict and explain the phenomena and mechanism of rain-wind-induced cable vibration than the single degree of freedom model. Nevertheless, the single degree of freedom model is acceptable to obtain the global maximum vibration response of the cylinder in respect of its simplicity. More detailed information on the 2DOF model can be found in [29].

It should be pointed out that the above analytical models of rain-wind-induced vibration did not consider the effects of axial flow and wind turbulence, so their effects on RWIV need further investigation. Moreover, many new investigations are emerging using either wind tunnel experiments [20,34,35], full-scale measurements [36], analytical methods [37] or CFD numerical simulation methods [38]. The new results shed light on the mechanism of RWIV of stay cables.

5.5 Prediction of Rain-Wind-Induced Cable Vibration

Although analytical models could predict major features of RWIV observed from field measurements and rain-wind tunnel tests, they are based on cylinder models without considering the variation of mean wind speed along a stay cable and the effect of mode shapes of cable vibration. The comparison between the analytical and field measurement results cannot be performed satisfactorily, and rain-wind tunnel test results alone cannot be used for full-scale stay cables. Furthermore, damping devices such as viscous dampers have been installed on stay cables near anchorages to mitigate RWIV, even though the mechanism of RWIV has not been fully understood. There is almost no analytical model available to evaluate the effectiveness of viscous dampers for vibration control of full-scale stay cables under rain-wind excitation.

This section thus presents an analytical model for investigating RWIV of stay cables in a cable-stayed bridge. The single degree of freedom (SDOF) rain-wind excitation model developed in Section 5.4 is first applied to a taut stay cable, taking into consideration the variation of mean wind speed along the stay cable and the effect of mode shapes of cable vibration. The computed results from the analytical model are then compared with available field measurement results. After a satisfactory comparison, parameter studies are performed to explore the mechanism of RWIV of full-scale stay cables.

5.5.1 Analytical Model for Full-Scale Stay Cables

Most stay cables in a cable-stayed bridge are of small sag in the order of 1% sag-to-length ratio, but with a high tension-to-weight ratio [3]. As a preliminary analytical study, the effects of sag and bending stiffness on cable vibration are neglected and only a taut inclined cable is considered and shown in Figure 5.12.

The inclination of the taut cable is denoted by angle α , and the yaw angle of incident wind is designated by angle β . The in-plane transverse motion $w(x, t)$ of a taut cable under rain-wind excitation can be described by the following partial differential equation of motion, which is essentially the same as Equation 5.1:

$$m \frac{\partial^2 w}{\partial t^2} - T \frac{\partial^2 w}{\partial x^2} + c \frac{\partial w}{\partial t} + F(x, t) = 0 \quad (5.42)$$

with the boundary conditions $w(x, t) = w(L, t) = 0$.

- In Equation 5.42:
- $w(x, t)$ is the in-plane transverse displacement normal to the cable axis at position x ;
 - m is the cable mass per unit length;
 - T is the cable tension force;
 - c is the internal cable damping coefficient;
 - $F(x, t)$ is the rain-wind induced force on the cable;
 - L is the length of the cable.

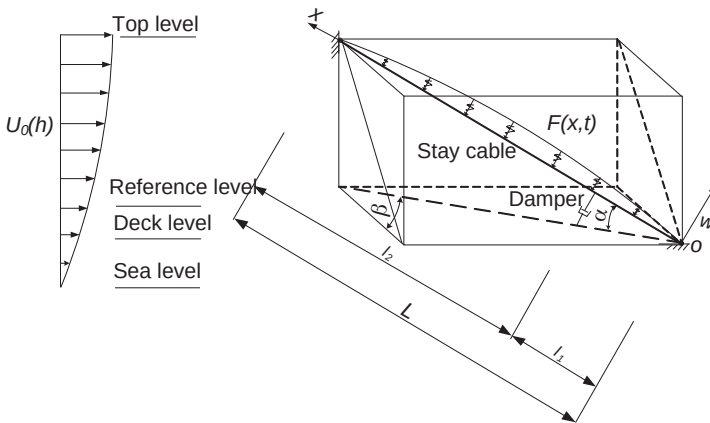


Figure 5.12 Orientation of rain-wind-induced vibration of a taut cable (from [46]) (Reproduced with permission from John Wiley & Sons Ltd).

By assuming that the upper rivulet vibrates circumferentially over the surface of the cable, rain-wind-induced force normal to the cable axis per unit length in the vertical direction, without considering turbulence and axis flow effects, can be expressed in terms of Equation 5.31:

$$F(x, t) = \frac{\rho D U_0(x)}{2} [R \dot{\theta}(x, t) \Gamma_1(x, t) + \dot{w}(x, t) \Gamma_2(x, t) + U_0(x) \theta(x, t) \Gamma_3(x, t)] \quad (5.43)$$

where:

R is the cable radius;

$\Gamma_1, \Gamma_2, \Gamma_3$ are the rain-wind aerodynamic functions.

The rain-wind aerodynamic functions not only include the function of cable inclination angle, wind yaw angle, wind stagnation point, and the mean wind speed via the static position of rivulet, but also the function of cable motion and rivulet motion, which may change the values of drag and lift coefficients as expressed by Equations 5.32 and 5.33.

In RWIV, the motion of upper rivulet is actually coupled with the motion of the cable. Therefore, the motion of upper rivulet should be taken as a variable and predicted together with the motion of cable in principle. This, however, requires a complicated analytical model for RWIV. Based on the observations from either field measurements or simulated rain-wind tunnel tests, the motion of upper rivulet, $\theta(x, t)$, is assumed to be harmonic and to follow the excited mode shape of the cable so far as a steady-state cable vibration is concerned.

$$\theta(x, t) = a \cdot W(x) \sin(\omega_r t) \quad (5.44)$$

where:

a is the maximum amplitude of rivulet motion, assumed to be a constant in this study;

$W(x)$ is the motion profile of rivulet along the cable axis, assumed to be the same as the excited mode shape of cable;

ω_r is the frequency of rivulet motion, assumed to be the same as the natural frequency of cable corresponding to the excited mode shape.

For a taut cable without damper, the motion profile and frequency of rivulet corresponding to the i th mode of cable vibration are given by:

$$W(x) = \sin\left(\frac{\pi i}{L} x\right) \quad (5.45)$$

$$\omega_i = i \omega_{o1} \quad (5.46)$$

where $\omega_{o1} = \frac{\pi}{L} \sqrt{\frac{T}{m}}$ is the first circular natural frequency of the taut cable.

For a stay cable in a cable-stayed bridge, mean wind speed actually varies along with cable axis. Mean wind speed is zero at the sea level, and it increases with height above the sea level in the atmospheric boundary layer as shown in Figure 5.12. The mean wind speed profile could be approximately expressed by a power law function:

$$U_0(h) = U_r \left(\frac{h}{h_r} \right)^{\alpha_1} \quad (5.47)$$

where:

$U_0(h)$ is the mean wind speed at height h above sea level;

U_r is the mean wind speed at a chosen reference height h_r ;

α_1 is the exponent depending on the surface roughness.

Since the height h can be expressed as a function of x along the cable axis, Equation 5.47 can be rewritten as:

$$U_0(x) = U_0(h(x)) = U_r \left(\frac{h_0}{h_r} + \frac{x \sin \alpha}{h_r} \right)^{\alpha_1} \quad (5.48)$$

where h_0 is the height of lower anchorage of the cable above the sea level.

Equation 5.42 is a strongly non-linear equation due to physical interaction between cable, rivulet, and wind. The rain-wind-induced force expressed in terms of Equation 5.43 is also an implicit function of cable motion and rivulet motion. The Runge-Kutta-Fehlberg method is therefore employed to find the numerical solutions [39]. This method is adaptive and easily implemented for the concerned numerical problem, in which a stay cable is divided into a number of sections along the cable axis and the numerical solutions are found at these sections for a series of time intervals using the relevant shape functions.

When the Runge-Kutta-Fehlberg method is used, only six evaluations are required per step, and local truncation error can be kept below a prescribed value by varying the step length. Note that rain-wind-induced force depends on static rivulet position and cable motion, while the static rivulet position varies with wind speed. Rain-wind-induced force on the cable needs to be computed section by section, starting from the static rivulet position and cable motion at the previous time step.

5.5.2 Prediction of Rain-Wind-Induced Vibration of Full-Scale Stay Cable

In this section, the analytical model and the numerical method proposed above are applied to a stay cable in a real cable-stayed bridge to compute its rain-wind-induced response and the computed response is compared with the measured one.

Stay cable No.14 of the Meikonishi West Bridge, as investigated by Hikami and Shiraishi [16] has been selected for this purpose. The length of the cable is 75 m; its diameter is 0.14 m; the mass per unit length is 51 kg/m; the tension force in the cable is 1.147×10^6 N; the first structural damping ratio of the cable is estimated at 0.0011; the inclination of the cable is about 45° ; and the height of lower anchorage of the cable above the sea level h_0 is 45 m.

The field measurement carried out by Hikami and Shiraishi [16] recorded that when cable No. 14 was down inclined along wind direction, with a wind yawed angle of about 45° , the simultaneous occurrence of wind and rain caused excessive vibration of the cable in the first mode of vibration, with a natural frequency about 1 Hz. They also conducted a series of wind tunnel tests, with rain conditions simulated, and reproduced rain-wind-induced cable vibration in their model tests. They observed from their model tests that the rivulet oscillated in a circumferential direction at the same period of the cable motion, and that the amplitude of rivulet motion was around 10° . The drag and lift coefficient curves of the cylinder with artificial upper rivulet, and the rivulet static position curve as the function of mean wind speed, were also given by Yamaguchi [25] and Hikami and Shiraishi [16], respectively. These are reproduced in Figures 5.8a and 5.8b respectively.

In addition to the information provided above, the power exponent α_1 in the wind profile is selected as 0.16 for an open fetch. The stagnation point influence factor ε is selected as 0.4. The wind reference height h_r is 47 m above the sea level and the mean wind speed at the reference height is taken as a variable. The cable is then divided into 50 sections along the cable axis. The basic parameters used in the simulation of rain-wind-induced cable vibration are listed in Table 5.1. The analytical model and the

Table 5.1 Basic parameters used in the simulation of rain-wind-induced cable vibration (from [46]) (Reproduced with permission from John Wiley & Sons Ltd)

Parameters	L (m)	D (m)	T ($\times 10^6$ N)	m (kg/m)	ζ_s	α ($^\circ$)	β ($^\circ$)	h_0 (m)	h_r (m)	ε	α_1
Values	75	0.14	1.147	51	0.11%	45	45	45	47	0.4	0.16

numerical method proposed are finally applied to the cable, with proper initial conditions and an error tolerance of 10^{-4} , to compute rain-wind-induced displacement response of the cable mode by mode.

Rain-wind-induced maximum displacement responses of cable No. 14, computed for the first three modes of vibration, are plotted in Figure 5.13 against different mean wind speeds at the reference level. The measured maximum displacement responses of the cables in the field are also presented in Figure 5.13, which includes cable No. 14 with its first two modes of vibration and cable No. 15 with its second and third modes of vibration [16].

Although rain-wind-induced cable vibration is a very complicated air-fluid-solid interaction problem, the computed results based on the proposed rain-wind-induced force model could capture the main features observed in the field. Both the computed and measured results demonstrate that rain-wind-induced cable vibration only occurs within a certain range of mean wind speed. It is also a kind of vibration of restricted amplitude other than galloping.

The predicted global maximum response is, however, larger than the measured one. This is because there exist uncertainties in both the measurement data and the modeling of rain-wind-induced cable vibration. For instance, the so-called measured maximum response was actually calculated from the response measured at a point 2 m above the deck level by assuming a sinusoidal mode shapes of cable vibration [16], the stagnation point influence factor used is approximate, and the influence of turbulent intensity and axial flow is neglected. Well-planned and carefully implemented full-scale measurements and wind tunnel tests are needed in the future in order to improve the quality of comparison and to refine the proposed model if necessary.

The computed displacement time history of the cable at its middle point in the first mode of vibration is displayed in Figure 5.14. It can be seen that rain-wind-induced response increases very quickly at the beginning, but the increasing rate becomes smaller and smaller. After about 300 seconds, the response amplitude remains constant, with the amplitude being about 0.22 m. The computed rain-wind-induced maximum displacement responses depicted in Figure 5.13 also manifest that for cable No. 14, the maximum displacement response in the first mode of vibration is largest while the maximum

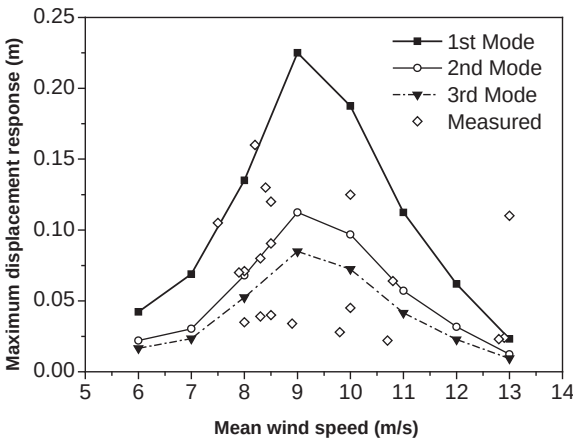
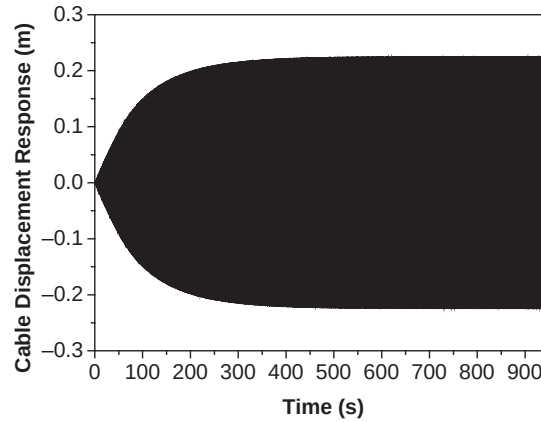
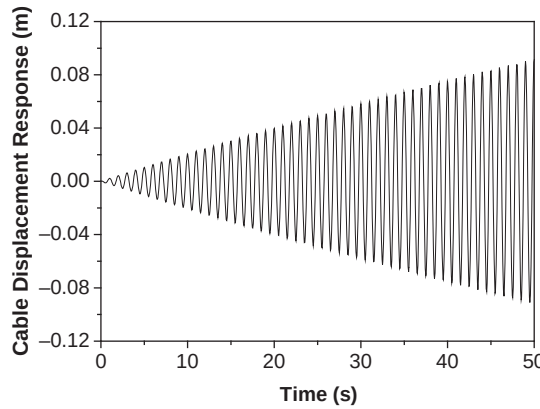


Figure 5.13 Computed and measured rain-wind-induced maximum displacement responses (from [46]) (Reproduced with permission from John Wiley & Sons Ltd).



(a) 900 seconds duration



(b) first 50 seconds

Figure 5.14 The global maximum displacement response time history of the cable in the first mode of vibration (from [46]) (Reproduced with permission from John Wiley & Sons Ltd).

displacement response in the third mode of vibration is at its smallest. However, the unfavorable mean wind speed range in which rain-wind-induced cable vibration occurs is almost the same for the first three modes of vibration.

5.5.3 Parameter Studies

To understand further the mechanisms of rain-wind-induced vibration of the stay cable, the effects of cable tension force, wind profile and cable internal damping are investigated. The investigation is performed by altering one parameter while keeping all other parameters unchanged.

Let us first consider the effect of cable tension on rain-wind-induced response of cable No. 14. While all the other parameters of the cable and the rivulet are kept the same as those used in the last section, the tension force of the cable is taken as a variable. The values of the tension force are selected as 0.765×10^6 ($2/3$ the original tension force T), 1.147×10^6 (the original tension force T) and 1.529×10^6 ($4/3$ the original tension force T).

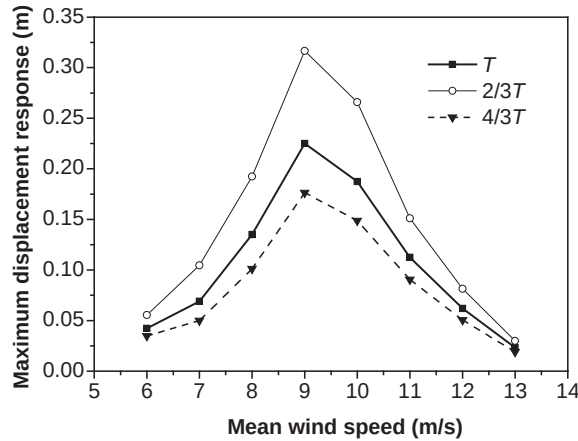


Figure 5.15 Effect of cable tension force on cable response in the first mode of vibration (from [46]) (Reproduced with permission from John Wiley & Sons Ltd).

Figure 5.15 shows rain-wind-induced maximum displacement response of the cable in the first mode of vibration against the mean wind speed at the reference level for three tension forces. It can be seen that the cable vibration remains velocity-restricted and amplitude-restricted features under three different tension forces. The tension force in the cable does not change the unfavorable mean wind speed range in which rain-wind-induced cable vibration occurs. The critical mean wind speed at which the global maximum displacement response of the cable occurs is almost the same under three different tension forces. However, as the tension force decreases (i.e. the cable becomes more flexible), the displacement responses of the cable increase. Similar results are found in the second and third modes of vibration, but with relatively small displacement responses.

Let us now consider the influence of mean wind speed profile on rain-wind-induced response of cable No. 14. The power exponent α_1 in the wind profile reflects the change of terrain. The increase in power exponent α_1 corresponds to the change of terrain from open sea to overland. While all the other parameters of the cable and the rivulet remain unchanged, including the reference level of mean wind speed, α_1 is taken as a variable and is selected as 0.16, 0.28 and 0.40 respectively.

Figure 5.16 shows the rain-wind-induced maximum displacement responses of the cable in the first and second modes of vibration against the mean wind speed at the reference level for three different power exponents. It can be seen that rain-wind-induced vibration of the cable remains velocity-restricted and amplitude-restricted for three different power exponents. However, as α_1 increases, the unfavorable wind speed range in which rain-wind-induced cable vibration occurs, and the critical mean wind speed at which the global maximum displacement response of the cable occurs, shift moderately to the lower wind speed side. This is because the change of power exponent of mean wind speed actually alters the mean wind speed which, in turn, changes the static position of upper rivulet and then the lift coefficient and, finally, the unfavorable wind speed range. On the other hand, the change in cable tension force does not affect the static position of rivulet and, therefore, it does not affect unfavorable wind speed range.

It can also be seen from Figure 5.16 that the global maximum displacement response of the cable at the critical mean wind speed also decreases as the power exponent increases. In addition to the effect of turbulence, one may say that rain-wind-induced cable vibration will be less if the terrain is overland.

The last parameter considered is the internal structural damping ratio in cable No. 14. While all other parameters of the cable and the rivulet are kept the same as before, the internal damping ratio is taken as a variable. For the first mode of vibration, the internal modal damping ratio is selected as 0.0011,

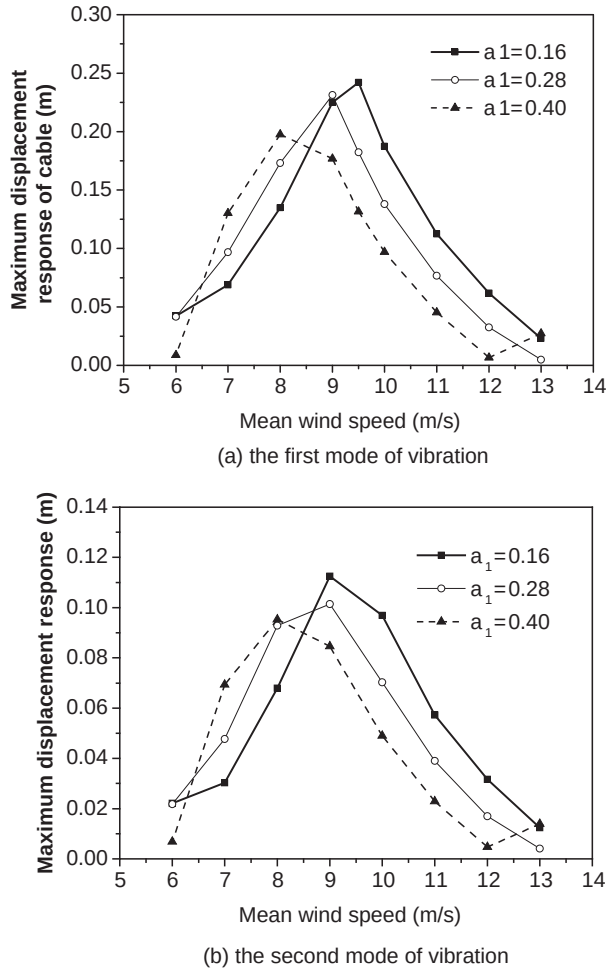


Figure 5.16 Effect of power exponent on cable response (from [46]) (Reproduced with permission from John Wiley & Sons Ltd).

0.0065 and 0.0111, and the computed rain-wind-induced maximum displacement responses of the cable are plotted in Figure 5.17 against the mean wind speed at the reference level. It can be seen that an increase of structural damping ratio to 0.65% can almost suppress rain-wind-induced cable vibration. Similar results are found for the second and third modes of vibration. It can thus be concluded that increasing cable damping is a very effective way to mitigate rain-wind-induced cable vibration. Rain-wind-induced vibration control of a stay cable will be discussed in Chapter 12.

5.6 Occurrence Probability of Rain-Wind-Induced Cable Vibration

To facilitate the engineering decision on whether or not RWIV will occur and anti-vibration measures of stay cables should be implemented, this section presents a framework for estimating the occurrence probability of RWIV in stay cables. The proposed framework is founded on the statistical analysis of wind speed, wind direction and rainfall intensity and the theoretical analysis of occurrence range of

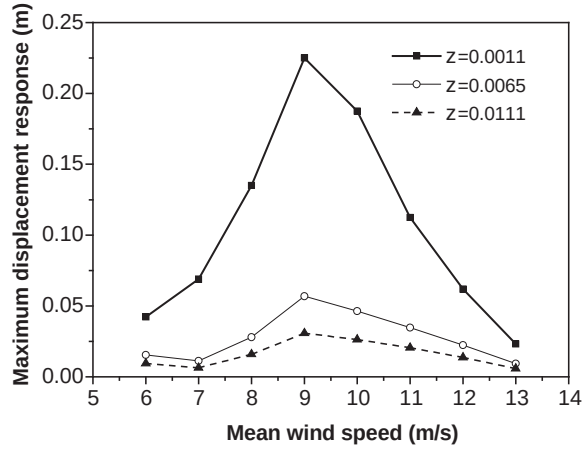


Figure 5.17 Effect of cable structural damping ratio on cable response in the first mode of vibration (from [46]) (Reproduced with permission from John Wiley & Sons Ltd).

RWIV in stay cables. The joint probability density function of wind speed and wind direction, and the probability density function of rainfall intensity are first obtained for the site of a bridge through statistical analysis. The occurrence range of wind speed and wind direction for RWIV in a stay cable greater than a vibration amplitude threshold value is then found based on a theoretical model described in Section 5.5. The occurrence probability and risk of RWIV are finally calculated according to the probability theory.

Figure 5.18 is a flow chart of the proposed framework, showing the estimation procedure for occurrence probability. The mathematical derivations and assumptions involved in each step are discussed in detail in subsequent sections. The practical use of the proposed framework is demonstrated in this section by taking a typical stay cable in a real cable-stayed bridge as a case study.

5.6.1 Joint Probability Density Function (JPDF) of Wind Speed and Direction

Excessive RWIV occurs mainly under low or moderate mean wind speeds. Therefore, a complete population of wind speed is of greater concern to this study than extreme values of wind speed. Various probability density functions (PDF) have been proposed to model a complete population of wind speed [40,41], but the convenience of the two-parameter Weibull distribution has encouraged its greater use than the other distributions [42]. With the lower limit being zero, the cumulative distribution function (CDF) and probability density function (PDF) of the Weibull form are, respectively:

$$P_u(U) = 1 - \exp \left[- \left(\frac{U}{c} \right)^k \right] \quad (5.49)$$

$$f_u(U) = \frac{k}{c} \left(\frac{U}{c} \right)^{k-1} \exp \left[- \left(\frac{U}{c} \right)^k \right] \quad (5.50)$$

where:

U is the wind speed;

$c(>0)$ is the scale parameter with the same unit as the wind speed;

$k(>0)$ is the shape parameter without dimension.

One possible weakness in the two-parameter Weibull distribution for wind speed is that it neglects the effect of wind direction [43,44]. Since excessive RWIV is closely related to wind direction, a joint

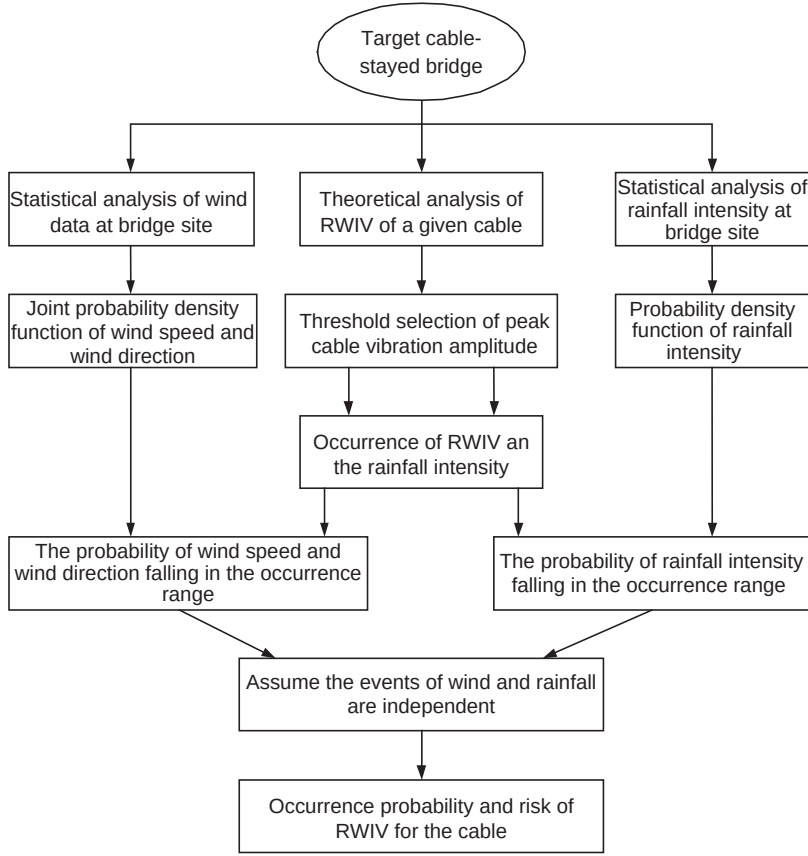


Figure 5.18 Flow chart of proposed probability-based framework (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

probability distribution of wind speed and wind direction has to be used for the study. To this end, a practical joint probability distribution function (see Chapter 2) is adopted in this study for a complete population of wind speed and wind direction based on two assumptions:

1. the distribution of the component of wind speed for any given wind direction follows the Weibull distribution;
2. the interdependence of wind distribution in different wind directions can be reflected by the relative frequency of occurrence of wind.

$$P_{u,\theta}(U, \theta) = P_{\theta}(\theta) \left\{ 1 - \exp \left[- \left(\frac{U}{c(\theta)} \right)^{k(\theta)} \right] \right\} = \iint f_{\theta}(\theta) f_{u,\theta}(U, k(\theta), c(\theta)) d\theta \quad (5.51a)$$

$$f_{u,\theta}(U, k(\theta), c(\theta)) = \frac{k(\theta)}{c(\theta)} \left(\frac{U}{c(\theta)} \right)^{k(\theta)-1} \exp \left[- \left(\frac{U}{c(\theta)} \right)^{k(\theta)} \right] \quad (5.51b)$$

$$P_{\theta}(\theta) = \int_0^{\theta} f_{\theta}(\theta) d\theta \quad (5.51c)$$

where:

$$0 \leq \theta \leq 2\pi;$$

$P_\theta(\theta)$ is the relative frequency of occurrence of wind in wind direction θ .

The occurrence frequency $P_\theta(\theta)$, as well as the distribution parameters $k(\theta)$ and $c(\theta)$, can be estimated using wind data recorded at the bridge site.

5.6.2 Probability Density Function of Rainfall Intensity

Rainfall intensity is another crucial factor for the onset of rain-wind-induced cable vibration. Either very light or very heavy rainfall could not provoke the relatively stable rivulet to be formed on the upper surface of a stay cable [19]. Statistical analysis of rainfall data at the bridge site is thus necessary to determine the probability distribution function of rainfall intensity, by which the probability of occurrence range of rainfall intensity that would give rise to cable vibration can be identified.

Suppose A and B denote, respectively, the rainy event and the rainy event that might induce cable vibration. The probability of event B is clearly the probability that both events A and B occur. This can be expressed as:

$$P_r(B) = P_r(A \cap B) = P_r(A)P_r(B|A) \quad (5.52)$$

The probability of event B can be determined by two ways: one is to calculate $P_r(B)$ by directly considering all the rainy events that might induce cable vibration; the other is to calculate the probability of event A and the conditional probability $P_r(B|A)$ to obtain $P_r(B)$. In this study the latter method is used, in order to be consistent with the method used by climatologists. To calculate the conditional probability $P_r(B|A)$, the probability density function of rainfall intensity should be established.

There are various probability density functions (PDF) proposed to model rainfall intensity [45]. The Gamma and Weibull distributions are explored in this study. The mathematical expression of the Gamma and Weibull probability density functions are, respectively, given by:

$$f_r(R) = \frac{1}{b^a \Gamma(a)} R^{a-1} e^{-\frac{R}{b}} \quad (5.53)$$

$$f_r(R) = \frac{d}{c} \left(\frac{R}{c} \right)^{d-1} e^{-\left(\frac{R}{c} \right)^d} \quad (5.54)$$

where:

$\Gamma(a)$ is the Gamma function $= \int_0^\infty y^{a-1} e^{-y} dy$;

R is the rainfall intensity;

a and b are the shape parameter and the scale parameter respectively for the Gamma distribution;

c and d are the shape parameter and the scale parameter, respectively, for the Weibull distribution.

5.6.3 Occurrence Range of Rain-Wind-Induced Cable Vibration

The occurrence range of wind speed and wind direction in which large-amplitude cable vibration will occur can be computed based on a recently-developed theoretical model presented in Section 5.5 [46].

Due to physical interaction between cable motion, rivulet motion and wind, the equation of motion of a taut cable under rain-wind excitation is non-linear. The Runge-Kutta-Fehlberg method is employed

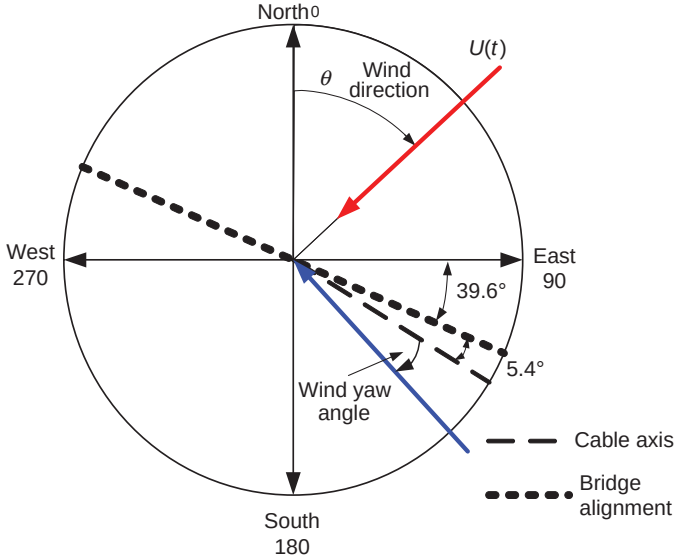


Figure 5.19 Definition of wind direction and wind yaw angle (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

to find the numerical solution. The peak vibration amplitude, $A(U, \beta)$, of a given stay cable with a given inclination under rain-wind excitation is computed for a series of mean wind speed, U , and wind yaw angle, β . It should be pointed out that β is presented with reference to the cable axis while the mean wind direction, θ , is counted with respect to the north in clockwise, as shown in Figure 5.19.

Thus, an angle transformation is necessary to transform $A(U, \beta)$ to $A(U, \theta)$ for a given stay cable in a given cable-stayed bridge. Once $A(U, \theta)$ is obtained, a threshold value A_s , beyond which RWIV is regarded unacceptable, can be selected. The occurrence range of wind speed and wind direction in which cable vibration amplitude exceeds the threshold value A_s can be determined and denoted as:

$$\Omega_{u,\theta}^{A_s} = \{U \in [U_1, U_2] \cap \theta \in [\theta_1, \theta_2] | A(U, \theta) \geq A_s\} \quad (5.55)$$

where U_1 , U_2 and θ_1 , θ_2 are the lower and upper bound of occurrence range of wind speed and wind direction, respectively.

5.6.4 Occurrence Probability of Rain-Wind-Induced Cable Vibration

The probability of wind speed and wind direction falling in the occurrence range $\Omega_{u,\theta}^{A_s}$ can be calculated as:

$$P_{u,\theta} \left\{ (U, \theta) \in \Omega_{u,\theta}^{A_s} \right\} = \iint_{\Omega_{u,\theta}^{A_s}} f_{\theta}(\theta) f_{u,\theta}(U, k(\theta), c(\theta)) du d\theta \quad (5.56)$$

Suppose Ω_r denotes the occurrence range of rainfall intensity for rain-wind-induced cable vibration. The probability of rainfall intensity falling in the occurrence range Ω_r can be calculated as $P_r(B|A) = P_r(B \in \Omega_r) = \int_{\Omega_r} f_r(R) dR$.

The occurrence probability of rain-wind-induced cable vibration with cable vibration amplitude larger than the threshold value can be finally calculated by:

$$P\{(U, \theta) \in \Omega_{u,\theta}^{A_s} \cap B \in \Omega_r\} = P_r(A)P_r(B \in \Omega_r)P_{u,\theta}\left((U, \theta) \in \Omega_{u,\theta}^{A_s} | B \in \Omega_r\right) \quad (5.57)$$

Assume that the two events of rain and wind occurrence are independent at this stage. Equation 5.57 can be reduced to:

$$P\{(U, \theta) \in \Omega_{u,\theta}^{A_s} \cap B \in \Omega_r\} = P_r(A)P_r(B \in \Omega_r)P_{u,\theta}\left((U, \theta) \in \Omega_{u,\theta}^{A_s}\right) \quad (5.58)$$

This assumption may be reasonably valid for monsoon situations in Hong Kong, in consideration that RWIV occurs mainly at low and moderate wind speeds, but it is not necessarily applicable to typhoon situations or other climates. Nevertheless, the framework proposed here can be extended without any considerable difficult if the correlation between the two events can be quantified in the future.

In terms of probability density functions of wind speed, wind direction and rainfall intensity, Equation 5.58 can be further written as:

$$P\{(U, \theta) \in \Omega_{u,\theta}^{A_s} \cap B \in \Omega_r\} = P_r(A) \int_{\Omega_r} f_r(R) dR \iint_{\Omega_{u,\theta}^{A_s}} f_\theta(\theta) f_{u,\theta}(U, k(\theta), c(\theta)) du d\theta \quad (5.59)$$

5.7 Case Study: Stonecutters Bridge

In this section, the proposed method is applied to the Stonecutters cable-stayed bridge to demonstrate its practical use. Hourly mean wind data and hourly rainfall data recorded near the bridge site are first analyzed to obtain the probabilistic models of both wind and rainfall intensity. By taking one typical cable as an example, the occurrence range of RWIV is then determined. The hourly occurrence probability and annual risk of rain-wind-induced vibration of the example cable are finally computed for different threshold values.

5.7.1 Statistical Analysis of Wind Data

There is a wind monitoring system consisting of four propeller anemometers and two ultrasonic anemometers set up nearby the concerned cable-stayed bridge. Wind records of hourly mean wind speed and wind direction within the period between 1 January 2000 to 31 December 2005 from the propeller anemometer are used in this study to find the joint probability density function of hourly mean wind speed and direction. The height of the propeller anemometer is 214 m above the sea level.

The wind records are carefully checked and abnormal records are eliminated first. Wind records having an hourly mean wind speed lower than 1 m/sec are then removed in order to avoid any adverse effect on the statistics. These qualified data are further divided into two categories – monsoon and typhoon – according to the records of the typhoon warning signal hoisted by the Hong Kong Observatory (HKO) during the period concerned. As a result, 19 775 hourly monsoon records are available for calculation of the joint probability density function of wind speed and direction.

All of the monsoon records are classified into 16 sectors of the compass, with an interval of $\Delta\theta = 22.5^\circ$ according to the hourly mean wind direction. In each sector, mean wind speed is further divided into 16 ranges, from zero to 32 m/sec, with an interval of $\Delta U = 2$ m/sec. This leads to a total of 256 cells, and the relative frequency of hourly mean wind speed and wind direction in each cell is listed in Table 5.2. The last row in Table 5.2 gives the relative frequency of mean wind speed without

Table 5.2 Relative frequencies of hourly mean wind speed and wind direction (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd)

Direction	0–2	2–4	4–6	6–8	8–10	10–12	12–14	14–16	16–18	18–20	20–22	22–24	24–26	26–28	28–30	30–32	SUM
N	0.081	0.465	0.126	0.152	0.081	0.071	0.076	0.106	0.071	0.035	0.020	0.005	0.010	0.000	0.005	0.000	1.305
NNE	0.101	0.753	0.647	0.647	0.440	0.324	0.314	0.197	0.142	0.066	0.046	0.015	0.000	0.000	0.000	0.000	3.692
NE	0.086	0.870	0.728	0.389	0.288	0.197	0.167	0.025	0.010	0.015	0.005	0.000	0.000	0.000	0.000	0.000	2.781
ENE	0.142	3.075	4.420	3.363	1.290	0.435	0.177	0.278	0.263	0.147	0.056	0.046	0.010	0.010	0.000	0.005	13.714
E	0.243	3.798	6.144	6.528	2.190	0.501	0.329	0.253	0.152	0.116	0.116	0.046	0.020	0.005	0.000	0.000	20.440
ESE	0.253	3.004	4.733	4.243	1.446	0.470	0.217	0.137	0.056	0.020	0.005	0.010	0.010	0.000	0.000	0.000	14.604
SE	0.293	2.908	3.965	2.655	0.860	0.430	0.142	0.051	0.040	0.025	0.005	0.010	0.000	0.000	0.000	0.005	11.388
SSE	0.172	1.760	2.528	2.215	0.753	0.354	0.121	0.091	0.051	0.010	0.010	0.005	0.005	0.000	0.000	0.000	8.076
S	0.283	1.350	2.336	2.599	0.925	0.465	0.228	0.086	0.025	0.025	0.005	0.015	0.010	0.005	0.000	0.000	8.359
SSW	0.187	0.910	1.163	1.077	0.531	0.142	0.167	0.137	0.106	0.025	0.000	0.005	0.000	0.000	0.000	0.000	4.450
SW	0.071	0.531	0.855	0.753	0.314	0.051	0.015	0.005	0.005	0.000	0.005	0.000	0.000	0.000	0.000	0.000	2.604
WSW	0.101	0.278	0.455	0.329	0.066	0.030	0.010	0.005	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.279
W	0.101	0.602	0.637	0.607	0.182	0.071	0.015	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	2.220
WNW	0.091	0.475	0.435	0.303	0.061	0.015	0.010	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.391
NW	0.051	0.379	0.142	0.147	0.051	0.035	0.000	0.000	0.000	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.809
NNW	0.308	1.102	0.622	0.389	0.223	0.131	0.025	0.035	0.030	0.005	0.010	0.000	0.005	0.000	0.000	0.000	2.887
SUM	2.564	22.260	29.937	26.397	9.699	3.722	2.013	1.411	0.956	0.496	0.283	0.157	0.071	0.020	0.005	0.010	100.0

considering wind direction, and the last column shows the relative frequency of mean wind direction without considering wind speed.

5.7.2 Joint Probability Density Function of Wind Speed and Wind Direction

Based on the relative frequencies of wind speed and wind direction calculated above, the theoretical expression of joint probability density function is deduced, based on Equation 5.51. The Weibull function is used to fit the histogram of hourly mean wind speed for each wind direction, and the typical results in the east, south and west directions are depicted in Figures 5.20a to 5.20c, respectively. The Weibull parameters identified for each wind direction are listed in Table 5.3, together with the relative frequency of mean wind direction. The coefficient of determination to measure the quality of fitting is also given in Table 5.3 for each wind direction.

It can be seen that, except for the north direction, the Weibull function fits wind data satisfactorily. The Weibull function is also applied to the complete wind records without considering wind direction, as shown in Figure 5.20d. The Weibull parameters identified for this case are

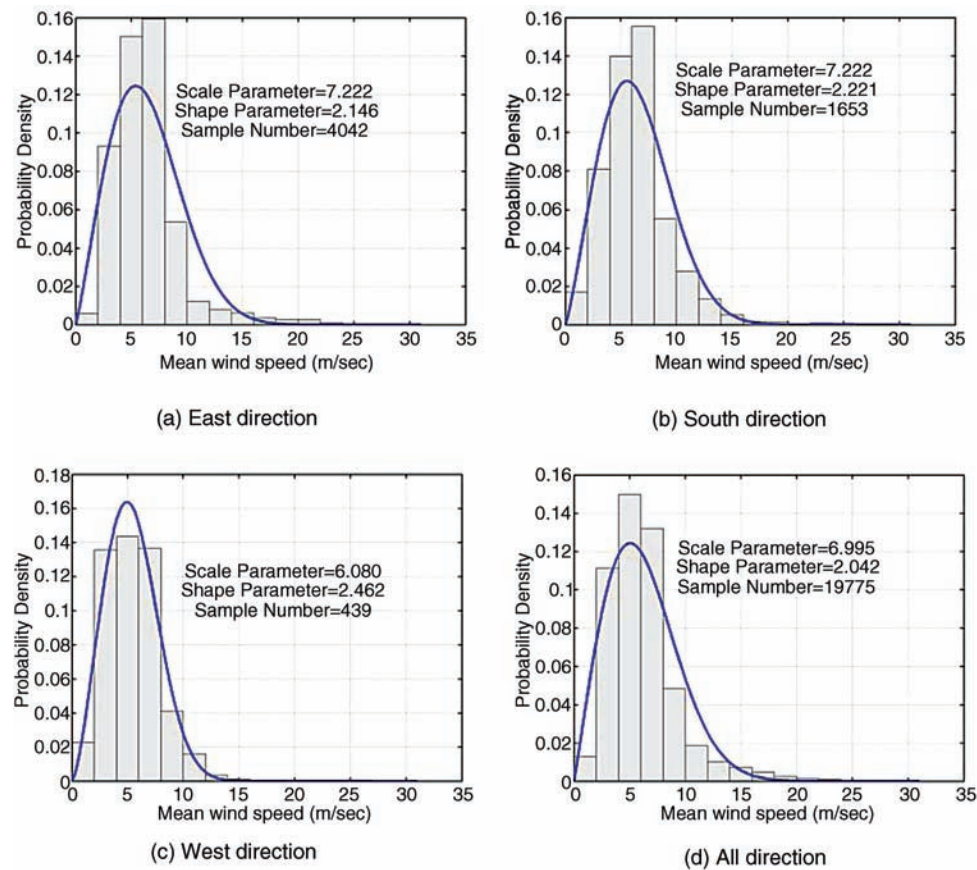


Figure 5.20 Weibull distribution and relative frequency of mean wind speed (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

Table 5.3 Identified parameters in different wind direction sectors (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd)

Direction	Record No.	$f_{\theta}(\theta)$	$c(\theta)$	$k(\theta)$	Coefficient of determination
N	258	0.013	8.501	1.394	0.6943
NNE	730	0.037	9.100	1.823	0.9165
NE	550	0.028	6.931	1.940	0.8914
ENE	2712	0.137	7.270	1.913	0.9826
E	4042	0.204	7.222	2.146	0.9840
ESE	2888	0.146	6.764	2.344	0.9929
SE	2252	0.114	6.384	2.215	0.9916
SSE	1597	0.081	6.819	2.242	0.9887
S	1653	0.084	7.266	2.221	0.9753
SSW	880	0.045	7.460	1.949	0.9723
SW	515	0.026	6.465	2.521	0.9987
WSW	253	0.013	5.904	2.295	0.9823
W	439	0.022	6.080	2.462	0.9771
WNW	275	0.014	5.392	2.438	0.9840
NW	160	0.008	5.454	1.992	0.8268
NNW	571	0.029	5.745	1.649	0.9248
Total	19 775	1.000	6.995	2.042	0.9869

given in the last row of Table 5.3. The results show that the Weibull function also fits the complete wind data adequately.

The relative frequency of wind direction and the scale and shape parameters of the Weibull function calculated above are also given in polar plot in Figures 5.21a to 5.21c, respectively.

It can be seen that the dominant monsoon direction is the east, and the scale and shape parameters do not vary significantly with wind direction. For the convenience of subsequent calculations, the data given in Table 5.3 and Figure 5.21, regarding the relative frequency of wind direction $f_{\theta}(\theta)$, the scale parameter $c(\theta)$ and the shape parameter $k(\theta)$, are fitted by the following harmonic functions [47]:

$$f_{\theta}(\theta) = a^f + \sum_{m=1}^{n_f} b_m^f \cos(m\theta - c_m^f) \tag{5.60a}$$

$$c(\theta) = a^c + \sum_{m=1}^{n_c} b_m^c \cos(m\theta - c_m^c) \tag{5.60b}$$

$$k(\theta) = a^k + \sum_{m=1}^{n_k} b_m^k \cos(m\theta - c_m^k) \tag{5.60c}$$

where:

a , b_m and c_m are the coefficients to be determined, whose superscripts f , c and k denote the relative frequency, the scale and shape parameters respectively;
 n_f , n_c and n_k are the order of harmonic functions.

Figures 5.22a to 5.22c display the histograms and the fitted harmonic functions for the relative frequency of wind direction, the scale and shape parameters, respectively. The corresponding coefficients identified for each function are listed in Table 5.4, together with the coefficient of determination. Five terms are used to fit the shape parameter and the relative frequency. For the scale parameter, only four terms are needed.

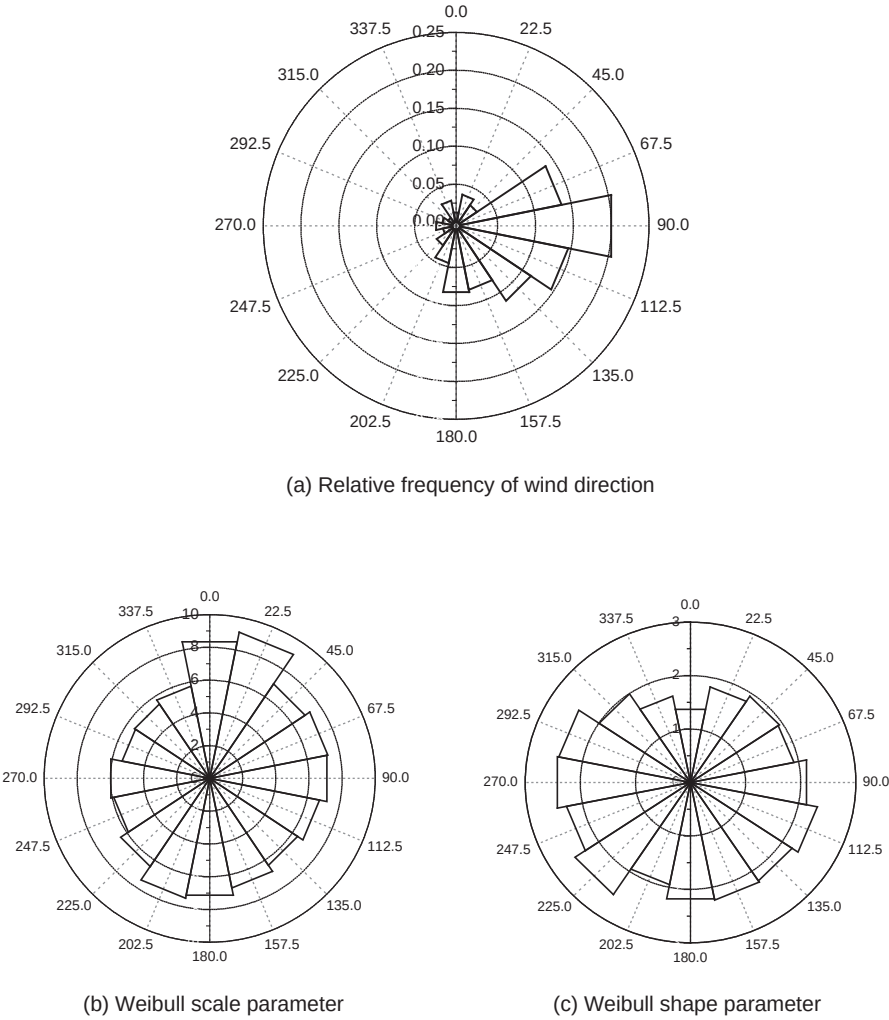


Figure 5.21 Relative frequency of wind direction and Weibull scale and shape parameters (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

5.7.3 Statistical Analysis of Rainfall Data

The sources of rainfall data used in this case study were acquired from the Hong Kong Observatory (HKO) and comprise nine years’ worth of hourly rainfall recordings from 1997 to 2005, recorded at the HKO station, which is fairly close to the concerned cable-stayed bridge. These statistics are considered to represent the general condition of rainfall in the vicinity of the bridge.

There are 78 888 available hourly rainfall records, which are accurate to 0.1 mm/hour. Any hourly rainfall less than 0.1 mm/hour is marked as trace rainfall in the collected records and is recorded as rainless in the analysis. As a result, the total of rainy records with hourly rainfall above 0.1 mm/hour comes to only 7385 over the nine years. Hourly rainfall is considered because a full development of rain-wind-induced cable vibration often needs more than half an hour, and this is also consistent with the hourly mean wind speed used in the statistical analysis of wind data.

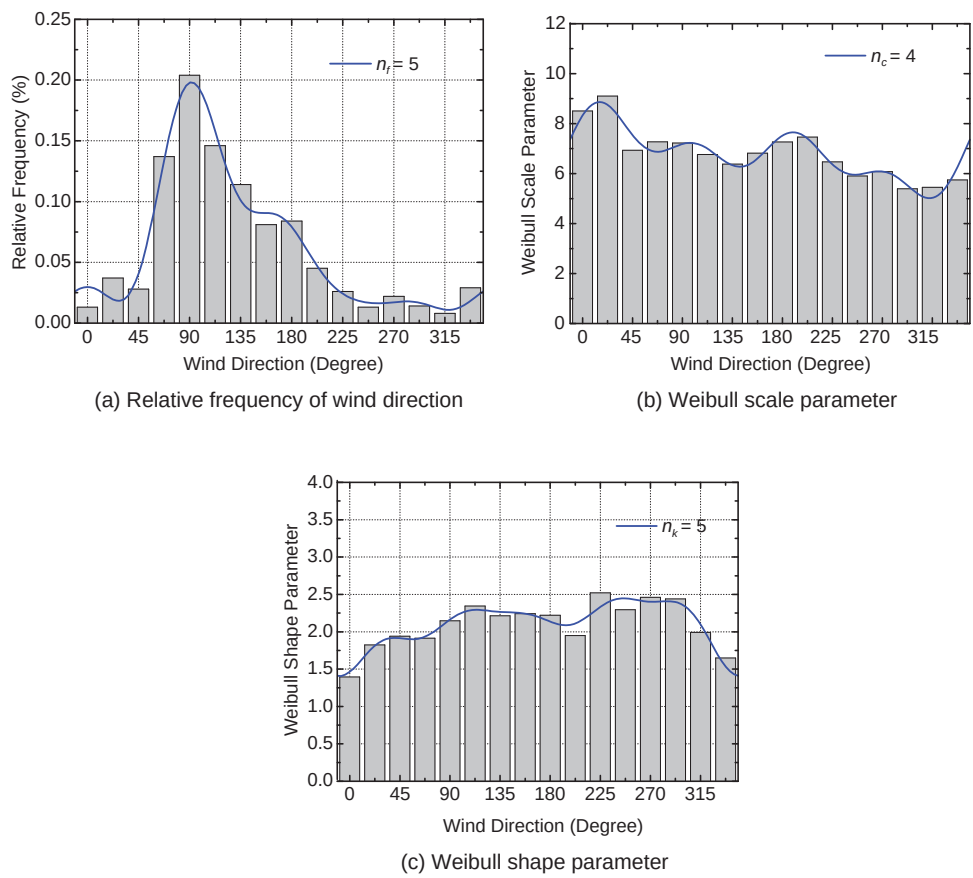


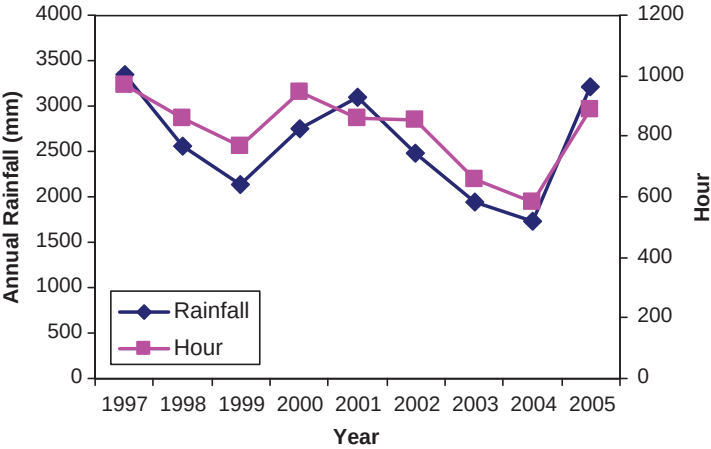
Figure 5.22 Histograms and fitted harmonic functions (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

Table 5.4 Identified coefficients in harmonic functions (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd)

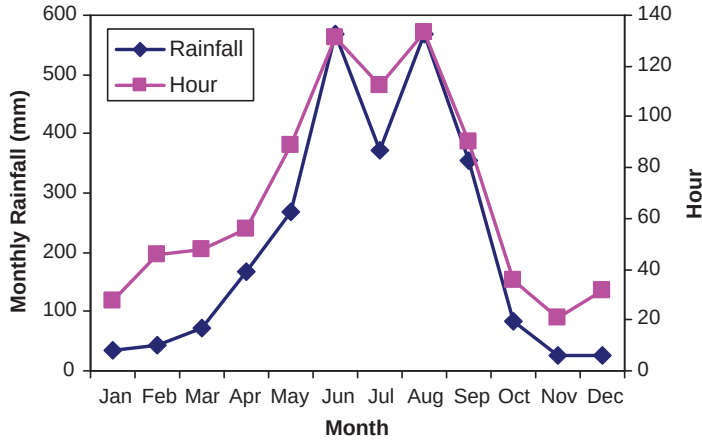
Coefficients	$f_{\theta}(\theta)$ $n_f = 5$	$c(\theta)$ $n_c = 4$	$k(\theta)$ $n_k = 5$
a	0.06 256	6.797	2.097
b ₁	0.06 882	0.7342	−0.2904
c ₁	1.949	1.325	0.3654
b ₂	−0.02 908	0.9061	−0.2432
c ₂	0.42	7.064	37.82
b ₃	−0.02 043	0.2764	0.09 093
c ₃	1.491	0.6311	1.947
b ₄	0.01 851	0.6069	0.07 955
c ₄	−0.1218	0.6603	−10.22
b ₅	0.00 645	—	0.07 557
c ₅	1.212	—	1.996
Coefficient of determination	0.9733	0.9157	0.9077

Let M and N ($M < N$) represent the total rainy hours and the total hours in the considered nine years, respectively. The relative frequency (probability) of rainy event $P_r(A) = M/N = 0.0936$. Figure 5.23a shows the annual rainfall and rainy hours from 1997 to 2005, based on the 7385 rainy records. Figure 5.23b depicts the average monthly rainfall and rainy hours.

It can be seen that, although the average monthly rainy hours vary over each month, the annual rainy hours remains relatively stable over the years. It can be also noted that the annual or average monthly rainfall is fairly correlated with the annual or average monthly rainy hours. To obtain the probability density function of hourly rainfall, the 7385 rainy records are classified according to their rainfall intensity into rainfall cells, starting from $R = 0.1$ mm/hour with an increment of $\Delta R = 0.25$ mm/hour. The resulting relative frequency of hourly rainfall is shown in Figures 5.24a and 5.25a.



(a) Annual rainfall and rainy hours



(b) Average monthly rainfall and rainy hours

Figure 5.23 Rainfall intensity and rainy hours (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

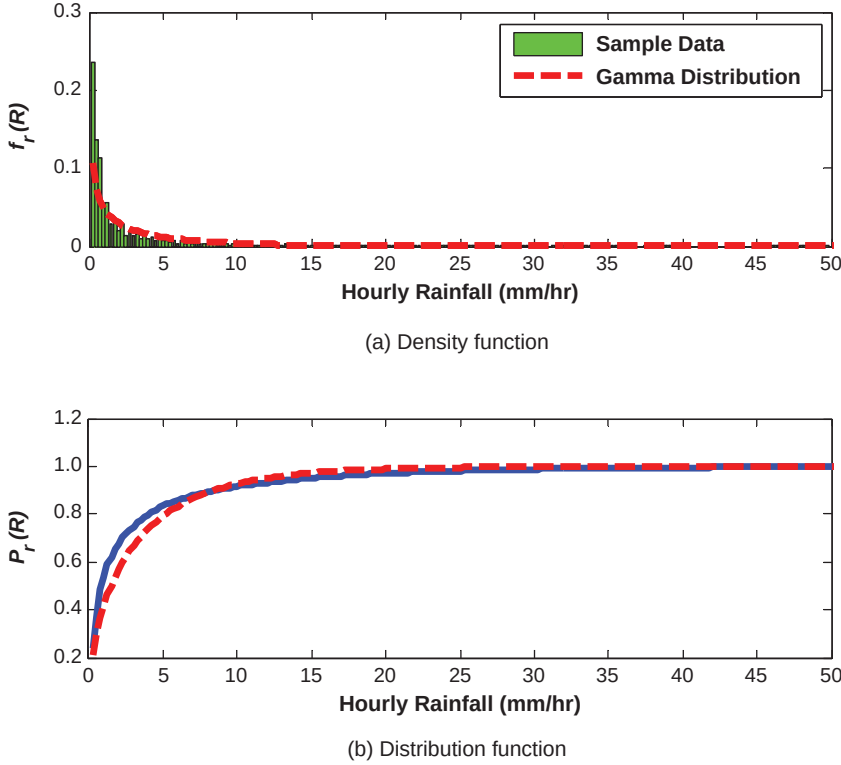


Figure 5.24 Gamma distribution for hourly rainfall (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

5.7.4 Probability Density Function of Rainfall Intensity

Both Gamma and Weibull distributions are adopted in this study to model the rainfall intensity. Figures 5.24a and 5.24b show the Gamma density function and the Gamma distribution function, respectively, together with the statistical results. Figures 5.25a and 5.25b display the Weibull density function and the Weibull distribution function, respectively, together with the statistical results. The estimated two parameters are $a=0.5415$ and $b=6.0406$ for the Gamma distribution expressed by Equation 5.53, and $c=2.0849$ and $d=0.6365$ for the Weibull distribution expressed by Equation 5.54. The coefficient of determination to measure the quality of fitting is 0.7432 for the Gamma distribution and 0.8192 for the Weibull distribution. The Weibull model is slightly better than the Gamma model in this case study.

By taking 3–30 mm/hour as the occurrence range Ω_r of hourly rainfall for rain-wind-induced cable vibration, the probability of rainfall intensity falling in the occurrence range Ω_r is calculated as $P_r(B \in \Omega_r|A) = P_r(B \in \Omega_r) = \int_{\Omega_r} f_r(R) dR = 0.2391$.

5.7.5 Occurrence Range of Rain-Wind-Induced Cable Vibration

After wind and rainfall data have been analyzed, a typical stay cable (No. 314) of the Stonecutters cable-stayed bridge is selected as a representative. The total length L , diameter D , tension force T , mass per length m , structural damping ratio in the first mode of vibration ζ_s , cable inclination α , and the first natural frequency $n_{01} (= \omega_{01}/2\pi)$ of the cable are listed in Table 5.5.

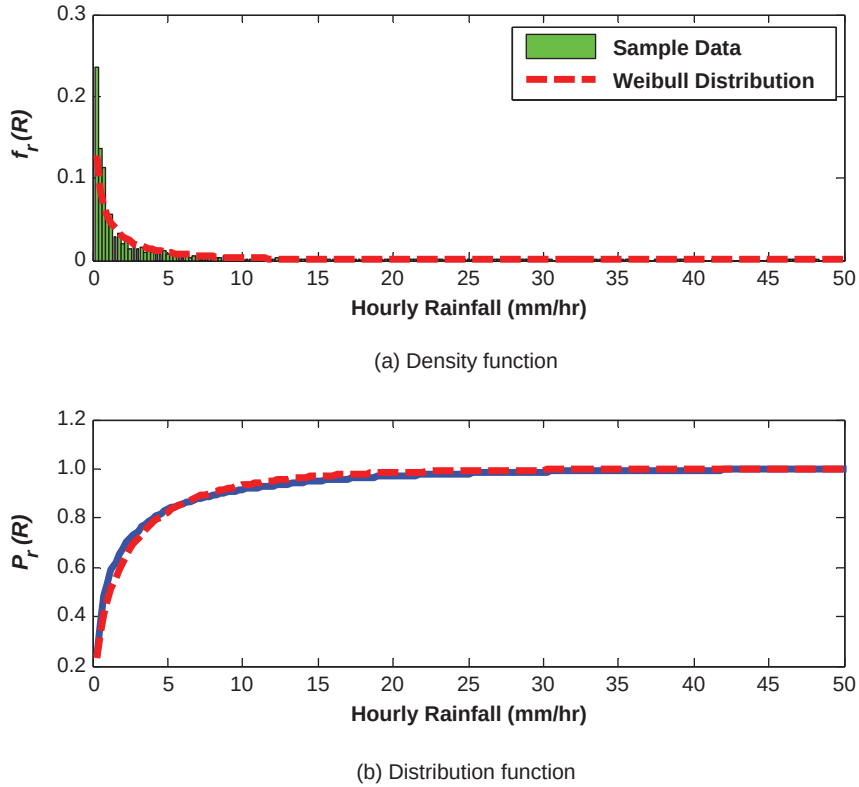


Figure 5.25 Weibull distribution for hourly rainfall (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd).

The cable is inclined downwards from southeast to northwest, as shown in Figure 5.19. The yaw wind from east or south, as shown in Figure 5.19, is considered in the determination of occurrence range of rain-wind-induced cable vibration. The bridge deck or the low anchorage of the cable is about 86 m above sea level. Since the bridge faces mainly an open-sea fetch from the south-west direction and mainly an over-land fetch from the north-east direction, two different mean wind speed profiles of power law form are considered. The power law exponent α_1 is 0.19 for the wind from the sea and 0.29 for the wind from the land. Wind reference height h_r is selected at the bridge deck level, i.e. 86 m above the sea level. The static position of upper rivulet is assumed to have the following expression for different yaw angle β , which is derived from the experiment results [20]:

$$\theta_0 = \frac{5550 \cdot \ln(0.045|\beta|)}{U^2} \quad 25^\circ \leq |\beta| \leq 85^\circ \quad (5.61)$$

Figure 5.26 shows the comparison of the static position of rivulet calculated using Equation 5.61 with the experiment results available in the literature. The aerodynamic functions, Γ_1 , Γ_2 , Γ_3 and the wind stagnation point influence factor ε suggested in the literature [27] are chosen in this case study.

Table 5.5 Main parameters of example stay cable used in case study (from [48]) (Reproduced with permission from Multi-Science Publishing Co Ltd)

Cable no.	L (m)	D (m)	T (ton)	m (kg/m)	ζ_s	α ($^\circ$)	n_{01} (Hz)
314	292.5	0.155	668.5	98.6	0.0015	30	0.44

Table 5.8 Hourly occurrence probability and annual risk of rain-wind-induced cable vibration (from [48])
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Threshold A_s	$0.5D$	$1D$	$2D$	$3D$
Probability ($\times 10^{-4}$)	18	11	6.17	1.98
Risk (%)	Almost 100%	Almost 100%	99.6	82.4

Rain-wind-induced cable vibration with mixed modes is not considered in this study. It can be seen from Tables 5.6 and 5.7 that the most severe cable vibration occurs when wind yaw angle β is between 35° and 55° and mean wind speed is between 7 m/s to 11 m/s, which are consistent with the field observations of rain-wind-induced vibration of stay cables in cable-stayed bridges.

5.7.6 Hourly Occurrence Probability and Annual Risk

Based on all the above results, the hourly occurrence probability of RWIV of the selected cable with cable vibration amplitude larger than a threshold value can be calculated using Equation 5.59. Four levels of vibration threshold A_s have been selected. They are $A_s \geq 0.5D$, $1D$, $2D$ and $3D$, where D is the diameter of the cable. The calculated hourly occurrence probability of RWIV of the cable for each threshold level is summarized in Table 5.8. It can be seen from this table that the hourly occurrence probability of RWIV of the cable is moderate.

For the selected cable, RWIV is about 16 hours, 10 hours, 5 hours and 2 hours in a year for the threshold $A_s = 0.5D$, $1D$, $2D$ and $3D$, respectively. The annual risk of RWIV of the cable is almost 100% for the threshold $A_s = 0.5D$ and $1D$, and 99.6% and 82.4% for the threshold $A_s = 2D$ and $3D$, respectively. It is noted that the cable selected is in the worst orientation for RWIV of all the stay cables.

The same procedure can be applied to other stay cables in the bridge, and the hourly occurrence probability and annual risk of all the stay cables can then be estimated. More detailed information on Sections 5.6 and 5.7 can be found in the literature [48].

5.8 Notations

- a (i) Amplitude of rivulet motion (Equation 5.35)
(ii) Shape parameter for Gamma distribution (Equation 5.53)
(iii) Coefficient (Equation 5.60)
- b (i) Scale parameter for Gamma distribution (Equation 5.53)
(ii) Coefficient (Equation 5.60)
- c (i) Internal cable damping coefficient (Equation 5.1)
(ii) Constant value (Equation 5.20)
(iii) Scale parameter for Weibull distribution (Equation 5.50)
(iv) Coefficient (Equation 5.60)
- c_r Damping coefficient of rivulet
- d (i) Sag inclination angle
(ii) Scale parameter for Weibull distribution (Equation 5.54)
- f Natural frequency of cable
- g Gravitational constant
- k Shape parameter (Equation 5.35)
- h_0 Height of lower anchorage of the cable above the sea level
- k_r Stiffness coefficient per unit length of the rivulet
- m Cable mass per unit length

m_r	Mass per unit length of rivulet
n	Order of harmonic function
s	Static equilibrium state of cable
u	Axial displacement
v	Out-of-plane transverse displacement
w	In-plane transverse displacement
$\dot{x}$	Relative velocity of the rivulet to the cylinder
ρ	Air density
α	Inclination angle
α_r	Predetermined exponent
α_1	Exponent depending on the surface roughness
ε	(i) Influence factor (ii) Dynamic strain
β	Yaw angle of incident wind
θ	(i) Motion of upper rivulet (Equation 5.35) (ii) Dynamic angular displacement of the rivulet as the cylinder vibrates
θ_0	Static position angle of upper rivulet due to mean wind
τ	Additional dynamic tension
γ	Ideal attack angle for the cylinder without rivulet
φ	Wind angle of attack
φ^*	Angle between relative velocity and the horizontal axis
ω_c	Circular natural frequency of cylinder
ω_r	Frequency of rivulet motion
ω_{01}	First circular natural frequency of taut cable
Γ	Rain-wind aerodynamic function
ϖ	Normalized circle frequency
ζ	Damping ratio of cable
ζ_a	Rain-wind aerodynamic damping ratio
ξ_c	Structural damping ratio of cylinder
ζ_s	Structural damping ratio
$\tilde{\xi}$	Total damping ratio of cylinder
λ^2	Sag parameter
A	Cross-section area of cable
A_i	Coefficients changing with the angle φ
A_n	Constant value (Equation 5.7)
A_{max}	Maximum peak to peak amplitude
A_{allow}	Allowable peak to peak amplitude
B_i	Coefficients changing with the angle φ
C_d	Drag coefficient
C_L	Lift coefficient
D	Cable diameter
E	Young's modulus of cable
L	Cable length
R	(i) Radius (Equation 5.24) (ii) Rainfall intensity (Equation 5.54)
S	Strouhal number
S_c	Scruton number
T	Cable tension force
U	Mean wind speed
$U_0(h)$	Mean wind speed at height h above the sea level

U_r	Mean wind speed at a chosen reference height h_r
U_{crit}	Minimum wind velocity for wake galloping instability
U_{rel}	Relative velocity of mean wind to the cylinder with moving rivulet
$f(t)$	Harmonic function of time with frequency ω
$f_u(U)$	Probability density function of Weibull distribution
$f_r(R)$	Mathematical expression of Gamma and Weibull probability density functions
$W(x)$	Motion profile of rivulet along the cable axis
$F(x, t)$	Wind-induced force on the cable
$\bar{F}(t)$	Normalized force due to the motion of rivulet
$P_\theta(\theta)$	Relative frequency of occurrence of wind in wind direction θ
$f_\theta(\theta)$	Relative frequency of wind direction
$P_u(U)$	Cumulative distribution function of Weibull distribution
$\Gamma(a)$	Gamma function (Equation 5.54)
$\Omega_{u,\theta}^{A_s}$	Occurrence range of wind speed and wind direction

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